CH3　應變分析

3-1. 正向應變與剪應變

一、正向應變

應變及浦松比

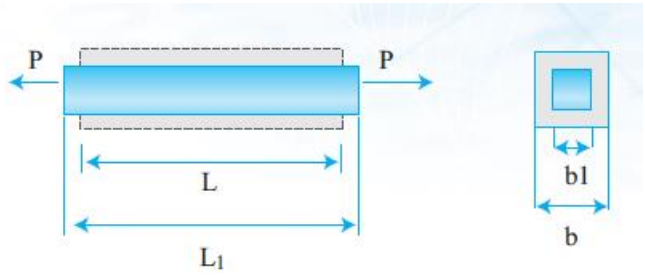
在材料力學當中，桿件不再假設為剛體，故當桿件受到軸向負載作用時，桿件若受拉力時則會伸長；若受壓力時則會縮短，桿件之變形是沿軸線方向的伸長和縮短，此一伸長或縮短之變形量，通常以符號 δ 表示，變形量 δ 與桿之原長度有關，將此一伸長或縮短之變形量 δ 與桿之原長L 之比值稱為應變(strain)亦即單位長度之變形量，以希臘字母 ε (epsilon)來表示，其說明如下所示：

(1)如圖　所示當桿件軸向拉伸（或壓縮）時，桿件會產生變形，其變形主要表現在沿軸向的伸長（或縮短），假設一等截面直桿原長爲L，橫截面面積爲A，在軸向拉力P的作用下，長度由L變爲L_1，桿件沿軸線方向的伸長爲 $\delta = L_1 - L$(拉伸時 δ 爲正，壓縮時 δ 爲負)。

(2)桿件的伸長量與桿的原長有關，將 δ 除以L，即以單位長度的伸長量來表徵桿件變形的程度，稱爲線應變，用 ε 表示：$\varepsilon = \frac{\delta}{L}$。

(3)徑向應變：在軸向力作用下，桿件沿軸向的伸長（縮短）的同時，徑向尺寸也將縮小（增大），假設橫向尺寸由b變爲$b_1 \Rightarrow \Delta b = b_1 - b$則徑向線應變爲 $\varepsilon' = \frac{\Delta b}{b}$。

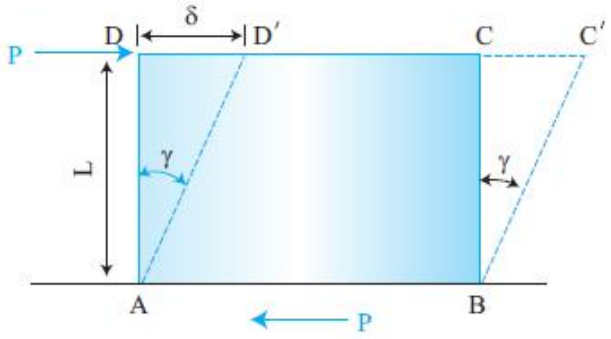
(4)浦松比：當桿件為同一種材料，其所受應力不超過比例極限時，徑向線應變與縱向線應變之比的絕對值爲常數，比值 ν 稱爲浦松比，即 $\nu = \left|\frac{\varepsilon'}{\varepsilon}\right|$，由於這兩個應變的符號恒相反，故有 $\varepsilon' = -\nu \cdot \varepsilon$。當軸向應變為正值時；橫向應變為負值，因此在上式中加一負號，使蒲松比值 ν 恆為正值，其值介於0～0.5 之間，一般金屬材料大約在1/4～1/3之間。



二、剪應變

由剪力所引起該元素扭曲的一種量度稱剪應變，也就是角度的改變，如圖所示，當物體承受剪力作用時，與剪力平行之平面，會發生相對平行之移動，當上端平面直接剪力V承受作用時，彈性材料會變形，物體受在水平方向由左而右的剪力，產生 δ 變形量，稱為橫向位移或總變形量，變形後產生 γ 角度的變化，此角度的變化 γ 即稱為剪應變，其單位為弧度(rad)，剪應變在介於平面間的角度減小，稱正剪應變，反之則稱為負剪應變。

如圖所示我們得知 $\tan\gamma=\frac{\delta}{L}$，但是因為剪應變很小，可視$\tan\gamma=\gamma$，因此可將剪應變改寫為 $\gamma=\frac{\delta}{L}$，根據虎克定律，即$\tau=G\gamma$，其中G稱之為剪割彈性係數(shear modulus of elasticity)或剛性係數(modulus of rigidity)，為一常數，依材料不同而有所不同，剪割彈性係數與彈性皆為材料的常數，其關係為 $G=\frac{E}{2(1+\upsilon)}$ (υ：浦松比)



3-2. 平面應變

一、應變分量

應變分量：僅剩ε_{xx}、ε_{yy}、γ_{xy}，以矩陣表示如下：

$$[\varepsilon]=\begin{bmatrix}\varepsilon_{xx} & \frac{\gamma_{xy}}{2}\\ \frac{\gamma_{xy}}{2} & \varepsilon_{yy}\end{bmatrix}=\begin{bmatrix}\varepsilon_{xx} & \varepsilon_{xy}\\ \varepsilon_{xy} & \varepsilon_{yy}\end{bmatrix}$$

其中ε_{xx}、ε_{yy}＝正應變(Normal strain)

γ_{xy}＝剪應變(Shear strain)

$\varepsilon_{xy}=\frac{\gamma_{xy}}{2}$＝剪應變之半

二、平面應變轉換

平面應變公式與平面應力公式相似，其要訣為將平面應力公式中σ轉換成ε，τ轉換成$\frac{\gamma}{2}$，

其相關公式如下所示:

(一)任意斜截面上的應變

$$\varepsilon_\theta=\frac{\varepsilon_x+\varepsilon_y}{2}+\frac{\varepsilon_x-\varepsilon_y}{2}\cos 2\theta+\frac{\gamma_{xy}}{2}\sin 2\theta$$

$$\frac{\gamma_\theta}{2}=-\frac{\varepsilon_x-\varepsilon_y}{2}\sin 2\theta+\frac{\gamma_{xy}}{2}\cos 2\theta$$

證明一:正向應變

考慮一邊長原為 dx 和 dy 的微小材料元素發生正向應變ε_x、ε_y(如圖(a)及圖(b)所示)，與剪應變γ_{xy}(如圖(c))，當三個應變(ε_x、ε_y、γ_{xy})同時發生時，則變形($\varepsilon_x dx$、$\varepsilon_y dy$、$\gamma_{xy} dy$)將分量投影到 x_1 軸可得累加變形為:

$$(\varepsilon_x dx+\gamma_{xy} dy)\cos\theta+\varepsilon_y dy\sin\theta$$

將變形除以原對角線之原長可得 x1 方向之應變ε_{x1}:

$$\varepsilon_{x1}=\frac{\varepsilon_x dx\cos\theta}{ds}+\frac{\gamma_{xy} dy\cos\theta}{ds}+\frac{\varepsilon_y dy\sin\theta}{ds}$$

由圖(a)可知 $\frac{dx}{ds}=\cos\theta$ 、 $\frac{dy}{ds}=\sin\theta$

$$\varepsilon_{x1}=\varepsilon_x\cos^2\theta+\gamma_{xy}\sin\theta\cos\theta+\varepsilon_y\sin^2\theta$$

$$\Rightarrow \varepsilon_{x_1} = \frac{\varepsilon_x + \varepsilon_y}{2} + \frac{\varepsilon_x - \varepsilon_y}{2}\cos 2\theta + \frac{\gamma_{xy}}{2}\sin 2\theta$$

若將上式 θ 以 θ +90° 代入可得

$$\varepsilon_{y1} = \frac{\varepsilon_x + \varepsilon_y}{2} - \frac{\varepsilon_x - \varepsilon_y}{2}\cos 2\theta - \frac{\gamma_{xy}}{2}\sin 2\theta$$

證明二:剪應變

考慮圖(d) 線段 $\overline{A'B} = \overline{A'C}\cos\theta - \overline{AC}\sin\theta$

$$\overline{A'B} = \varepsilon_y dy\cos\theta - (\varepsilon_x dx + \gamma_{xy} dy)\sin\theta$$

其中 $\beta = \frac{\overline{A'B}}{ds} = \frac{\varepsilon_y dx\cos\theta}{ds} - \frac{\gamma_{xy} dy\sin\theta}{ds} - \frac{\varepsilon_x dy\sin\theta}{ds}$

由圖(a)可知 $\frac{dx}{ds} = \cos\theta$ 、 $\frac{dy}{ds} = \sin\theta$

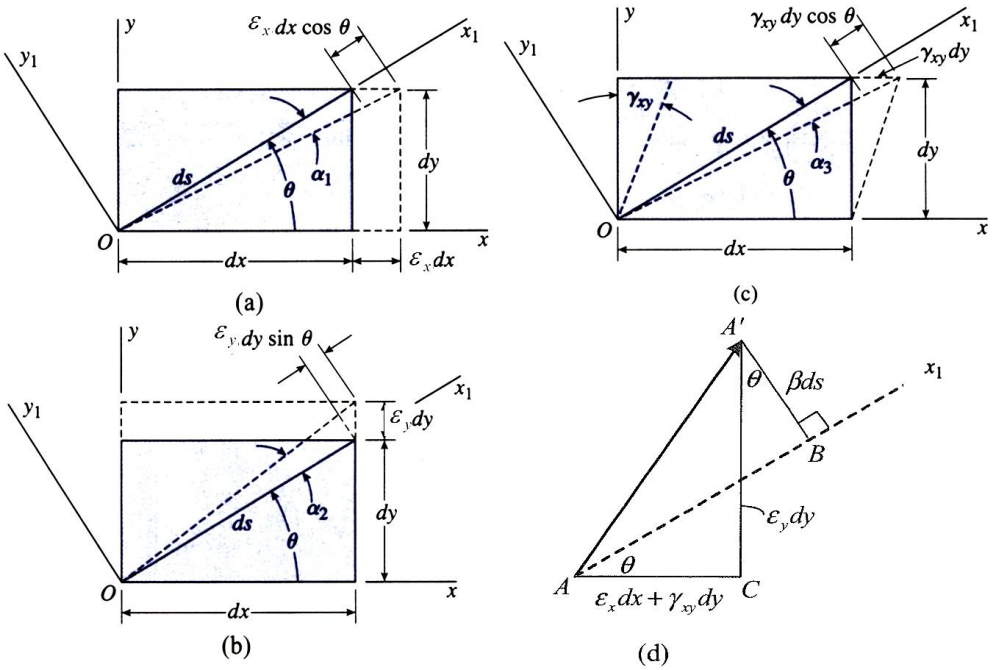
$$\beta = \varepsilon_y \sin\theta\cos\theta - \varepsilon_x \sin\theta\cos\theta - \gamma_{xy}\sin^2\theta$$

若將上式 θ 以 θ +90° 代入可得

$$\beta' = -\varepsilon_y \sin\theta\cos\theta + \varepsilon_x \sin\theta\cos\theta - \gamma_{xy}\cos^2\theta$$

$$\gamma_{x1y1} = \beta - \beta' = -2(\varepsilon_x - \varepsilon_y)\sin\theta\cos\theta + \gamma_{xy}(\cos^2\theta + \sin^2\theta)$$

$$\frac{\gamma_{x1y1}}{2} = -\frac{\varepsilon_x - \varepsilon_y}{2}\sin 2\theta + \frac{\gamma_{xy}}{2}\cos 2\theta$$



三、最大平面應變及最大剪應變

(一)最大主應變及方向

1．最大主應變：$\varepsilon_{1,2}=\dfrac{\varepsilon_x+\varepsilon_y}{2}\pm\sqrt{(\dfrac{\varepsilon_x-\varepsilon_y}{2})^2+(\dfrac{\gamma_{xy}}{2})^2}$

2．主軸方向的角度：$\tan(2\theta_p)=\dfrac{\gamma_{xy}}{\varepsilon_x-\varepsilon_y}$

證明:

最大主應變係為剪應變分量為零的斜面

$$\frac{\gamma_{xy}}{2}=-\frac{\varepsilon_x-\varepsilon_y}{2}\sin 2\theta+\frac{\gamma_{xy}}{2}\cos 2\theta$$

$\tan(2\theta_p)=\dfrac{\gamma_{xy}}{\varepsilon_x-\varepsilon_y}\Rightarrow$ 有兩根(θ_{p1}及θ_{p2})分別為最大主應變面及最小主應變面與x面的夾角

$$\theta_{p2}=\theta_{p1}+90^\circ$$

由上式可得

$$\sin(2\theta_{p1})=\frac{\gamma_{xy}}{\sqrt{(\varepsilon_x-\varepsilon_y)^2+(\dfrac{\gamma_{xy}}{2})^2}}$$

$$\cos(2\theta_{p1})=\frac{\varepsilon_x-\varepsilon_y}{\sqrt{(\varepsilon_x-\varepsilon_y)^2+(\dfrac{\gamma_{xy}}{2})^2}}$$

代回

$$\varepsilon_1 = \frac{\varepsilon_x + \varepsilon_y}{2} + \frac{\varepsilon_x - \varepsilon_y}{2}\cos 2\theta_{p1} + \frac{\gamma_{xy}}{2}\sin 2\theta_{p1}$$

可得

$$\varepsilon_1 = \frac{\varepsilon_x + \varepsilon_y}{2} + \sqrt{(\frac{\varepsilon_x - \varepsilon_y}{2})^2 + (\frac{\gamma_{xy}}{2})^2}$$

同理

$$\varepsilon_2 = \frac{\varepsilon_x + \varepsilon_y}{2} + \frac{\varepsilon_x - \varepsilon_y}{2}\cos 2\theta_{p2} + \frac{\gamma_{xy}}{2}\sin 2\theta_{p2}$$

$$\varepsilon_2 = \frac{\varepsilon_x + \varepsilon_y}{2} - \sqrt{(\frac{\varepsilon_x - \varepsilon_y}{2})^2 + (\frac{\gamma_{xy}}{2})^2}$$

故最大主應變**:** $\varepsilon_{1,2} = \frac{\varepsilon_x + \varepsilon_y}{2} \pm \sqrt{(\frac{\varepsilon_x - \varepsilon_y}{2})^2 + (\frac{\gamma_{xy}}{2})^2}$

(二)最大剪應變及方向

欲求最大剪應變，不可直接令 ε_θ 為零，因為最大剪應面上正應變不為零

1. 最大剪應變**:** $\boxed{\frac{\gamma_{\max}}{2} = \sqrt{(\frac{\varepsilon_x - \varepsilon_y}{2})^2 + (\frac{\gamma_{xy}}{2})^2}}$

2. 最大剪應變的方向角**:** $\tan(2\theta_s) = -\frac{\varepsilon_y - \varepsilon_x}{\gamma_{xy}}$

證明:

$$\frac{\partial \gamma_{xy}}{\partial \theta} = \frac{\partial}{\partial \theta}[2\times(-\frac{\varepsilon_x - \varepsilon_y}{2}\sin 2\theta + \frac{\gamma_{xy}}{2}\cos 2\theta)] = 0$$

$$\Rightarrow 2(\varepsilon_x - \varepsilon_y)\cos 2\theta + 2\gamma_{xy}\sin 2\theta = 0$$

$$\Rightarrow \tan(2\theta_s) = -\frac{\varepsilon_y - \varepsilon_x}{\gamma_{xy}}$$

$\tan(2\theta_s) = -\frac{\varepsilon_x - \varepsilon_y}{\gamma_{xy}} \Rightarrow$ 有兩根(θ_{s1}及$\alpha\theta_{s2}$)分別為最大剪應變面及最小剪應變面與x面的夾角

$$\theta_{s2} = \theta_{s1} + 90^\circ$$

由上式可得

$$\cos(2\theta_{s1}) = -\frac{\gamma_{xy}}{\sqrt{(\varepsilon_x - \varepsilon_y)^2 + (\frac{\gamma_{xy}}{2})^2}}$$

$$\sin(2\theta_{s1}) = \frac{\varepsilon_x - \varepsilon_y}{\sqrt{(\varepsilon_x - \varepsilon_y)^2 + (\frac{\gamma_{xy}}{2})^2}}$$

代回

$$\varepsilon_1 = \frac{\varepsilon_x + \varepsilon_y}{2} + \frac{\varepsilon_x - \varepsilon_y}{2}\cos 2\theta_{s1} + \frac{\gamma_{xy}}{2}\sin 2\theta_{s1}$$

$$\varepsilon_1 = \frac{\varepsilon_x + \varepsilon_y}{2}$$

$$\frac{\gamma_{\theta s1}}{2} = \frac{\gamma_{\max}}{2} = \sqrt{(\frac{\varepsilon_x - \varepsilon_y}{2})^2 + (\frac{\gamma_{xy}}{2})^2}$$

同理可證

$$\frac{\gamma_1}{2} = \frac{\gamma_{\min}}{2} = -\sqrt{(\frac{\varepsilon_x - \varepsilon_y}{2})^2 + (\frac{\gamma_{xy}}{2})^2}$$

故

$$\gamma_{\max} = \varepsilon_1 - \varepsilon_2$$

四、 應變之莫爾圓

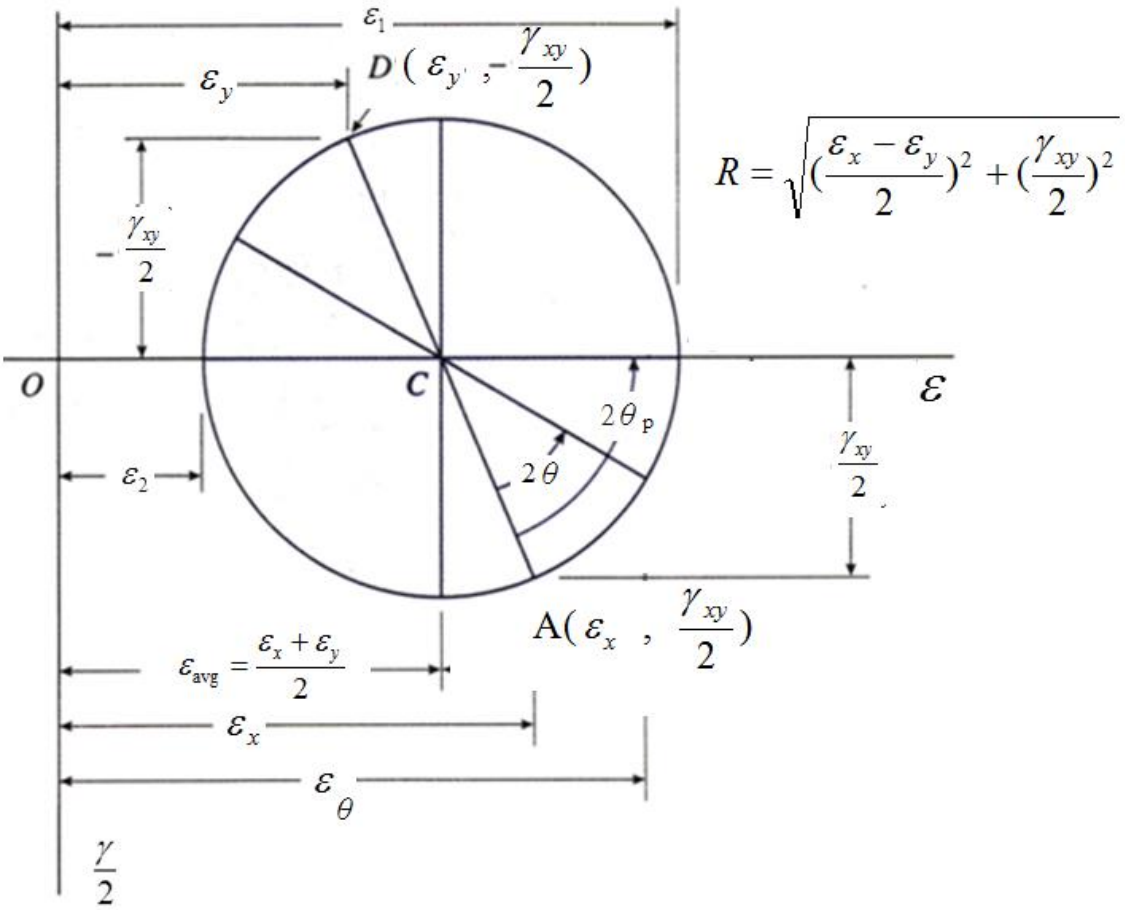
(一)平面應變之莫爾圓

1. 因平面應變轉換方程式數學上相似於平面應力轉換方程式，亦可利用莫爾圓求解應變轉換之問題，此法具有以圖解方式瞭解在某一點上之正向及剪應變分量如何從一元素方位變化至下一個方位。

$$(\varepsilon_x - \varepsilon_{avg})^2 + \left(\frac{\gamma_{xy}}{2}\right)^2 = R^2$$

2. 建立一座標統使得橫座標表示正向應變ε，向右為正，而縱座標表示剪應變值之一半，$\frac{\gamma}{2}$，向下為正。
3. 利用ε_x、 ε_y、γ_{xy}之正的慣用符號，如圖所示，定出圓心，其位於 x 軸上距原點 $\varepsilon_{avg} = \frac{\varepsilon_x + \varepsilon_y}{2}$ 。
4. 標上參考點 A 其具座標 A(ε_x ， $\frac{\gamma_{xy}}{2}$)，連接點 A 及圓心 C，此距離為此圓之半徑 $R = \sqrt{(\frac{\varepsilon_x - \varepsilon_y}{2})^2 + (\frac{\gamma_{xy}}{2})^2}$ ，而 CA 稱為徑向參考線，一旦定出 R，則可描繪此圓。
5. 應變元素轉θ角，則莫爾圓需轉 2θ角。
6. 將莫爾圓轉至主應變狀態可以順時針或逆時針，實際上以所需轉角小者為佳。
7. 如圖所示，若需轉到主應變狀態，應該將 A 點(面)逆時針轉$2\theta_p$；而若需轉到最大剪應

力狀態，則將 A 點(面)逆時針轉 $2\theta_p + 90^\circ$，在最大主應變平面上，只有正交應變而無剪應變，然而在最大剪應變平面，通常有正交應變，且正交應變值相同。

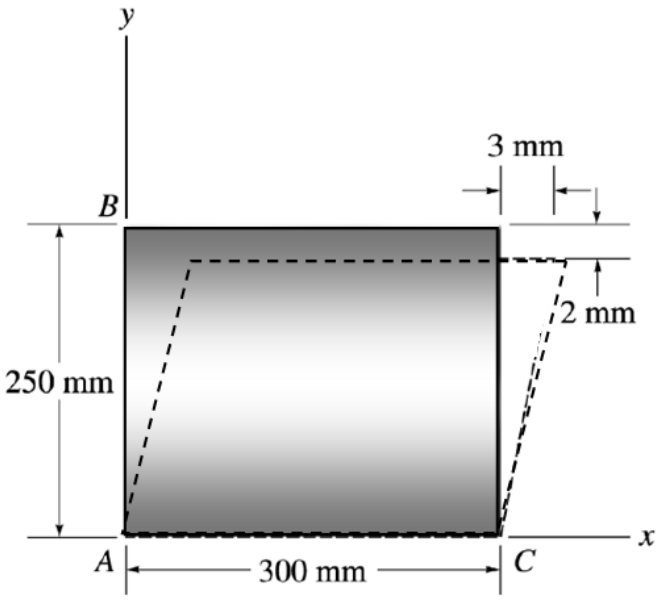


(二)電阻應變規(electrical-resistance strain gauge)

電阻應變規 (electrical-resistance strain gauge) 可用來量測拉伸試驗中試桿之正向應變，其係由金屬線格子或金屬箔黏貼在試桿上而構成。對於物體上一般負載，在其自由表面上某點之正向應變通常利用一組具三個電阻應變規，安置在一特定模式下而定出。此模式稱之為應變菊花座 (strain rosettes)。值得注意，這些應變僅在同平面之量規下作量測，而且因物體在垂直於量規方向係無應力的，故量規仍承受平面應力而不是平面應變。

範例 1

The plate is deformed into the dashed shape shown in Fig. If in this deformed shape horizontal lines on the plate remain horizontal and do not change their length, determine (a) the average normal strain along the side AB, and (b) the average shear strain in the plate relative to the x and y axes.

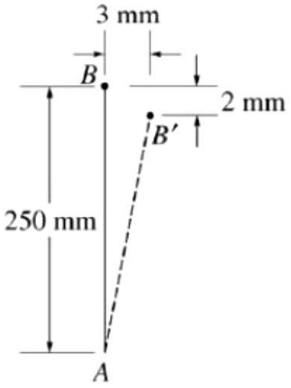


【104 中央土木】

【中文說明】

、圖一所示的平板受力發生變形，請計算變形後(點線所示)：(a) AB 線的平均正向應變(average normal strain) 及 (b) A 點相對於 x，y 軸的平均剪應變(average shear strain)　　(每小題各 10 分，共 20 分)

【解析】

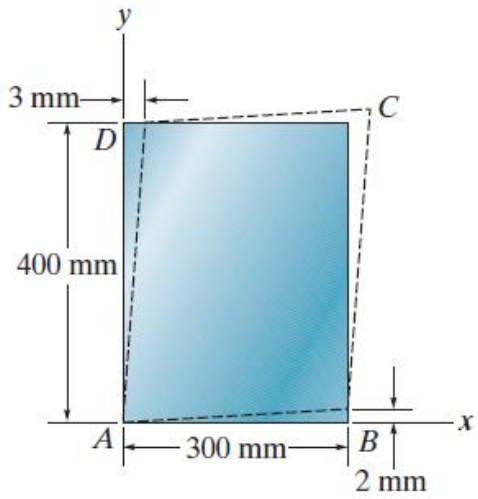


$$AB' = \sqrt{(250-2)^2 + 3^2} = 248.018(mm) \Rightarrow \varepsilon_{AB} = \frac{\sqrt{(250-2)^2 + 3^2} - 250}{250} = 0.00793$$

$$\gamma_{xy} = \tan^{-1}\left(\frac{3}{250-2}\right) = 0.0121(rad)$$

範例 2

The piece of rubber is originally rectangular.(a) Determine the average shear strain γ_{xy} at A if the corners B and D are subjected to the displacements that cause the rubber to distort as shown by the dashed lines.(b) Determine the average normal strain along the side AD.



【海洋機械】

【中文說明】

一平板變形如虛線所示，(a)試求平板相對於AB及AD線之剪應變(b)試求AD邊的正向應變

【解析】

(a)

$$\theta_1 = \tan\theta_1 = \frac{2}{300} = 0.006667 \text{ rad}$$

$$\theta_2 = \tan\theta_2 = \frac{3}{400} = 0.0075 \text{ rad}$$

$$\gamma_{xy} = \theta_1 + \theta_2$$

$$= 0.006667 + 0.0075 = 0.0142 \text{ rad}$$

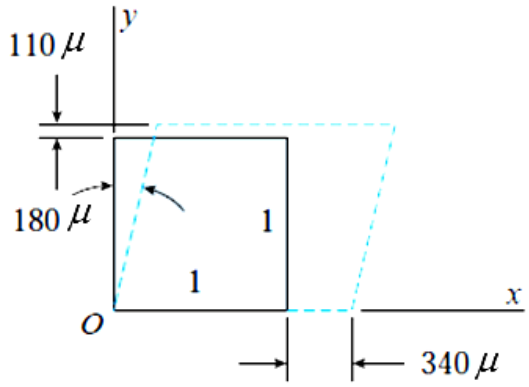
(b)

$$AD' = \sqrt{(400)^2 + (3)^2} = 400.01125 \text{ mm}$$

$$\varepsilon_{AD} = \frac{400.01125 - 400}{400} = 0.0281(10^{-3}) \text{ mm/mm}$$

範例 3

An element of material in plane strain undergoes the following strains: $\varepsilon_x = 340\mu$, $\varepsilon_y = 110\mu$, and $\gamma_{xy} = 180\mu$ These strains are shown highly exaggerated in Fig., which shows the deformations of an element of unit dimensions. Since the edges of the element have unit lengths, the changes in linear dimensions have the same magnitudes as the normal strains ε_x and ε_y. The shear strain γ_{xy} is the decrease in angle at the lower-left corner of the element. Determine the following quantities: (a) the strains for an element oriented at an angle $\theta = 30^\circ$, (b) the principal strains, and (c) the maximum shear strains. (Consider only the in-plane strains, and show all results on sketches of properly oriented elements.)



【成大機械】

【解析】

受平面上應變之元素，有如下的應變。 $\epsilon_x = 340\times10^{6}$ $\epsilon_v = 110\times10^{-6}$ $\gamma_{xv} = 180\times10^{-6}$

圖示顯示出單位大小之一元素的變形。由於元素的邊緣具單位強度，線尺寸的改變與正向應變 ϵ_x 及 ϵ_y 有相同的大小。在元素左下隅，剪應變 γ_{xv} 的角度減小了。

求如下量：(a) 朝向 30° 之元素的應變，(b) 主應變，及 (c) 最大剪應變。(只考慮平面上應變，並在適當指向的元素上示出所有的結果)。

【解析】

(a)

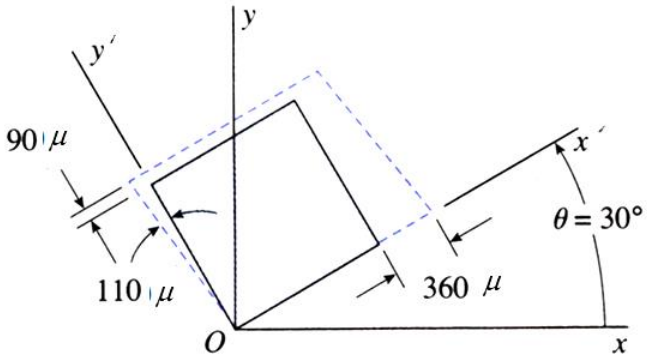
$$\varepsilon_\theta = \varepsilon_{x'} = \frac{\varepsilon_x + \varepsilon_y}{2} + \frac{\varepsilon_x - \varepsilon_y}{2}\cos 2\theta + \frac{\gamma_{xy}}{2}\sin 2\theta$$

$$\Rightarrow \frac{340+110}{2} + \frac{340-(110)}{2}\cos(2\times30^\circ) + \frac{180}{2}\sin(2\times30^\circ) = 360\mu$$

$$\frac{\gamma_\theta}{2} = -\frac{\varepsilon_x - \varepsilon_y}{2}\sin 2\theta + \frac{\gamma_{xy}}{2}\cos 2\theta$$

$$\Rightarrow -\frac{340-(110)}{2}\sin(2\times 30^\circ)+\frac{180}{2}\cos(2\times 30^\circ)=-55\mu$$

$$\varepsilon_x+\varepsilon_y=\varepsilon_{x'}+\varepsilon_{y'}\Rightarrow \varepsilon_{y'}=340\mu+110\mu-360\mu=90\mu$$



(b)

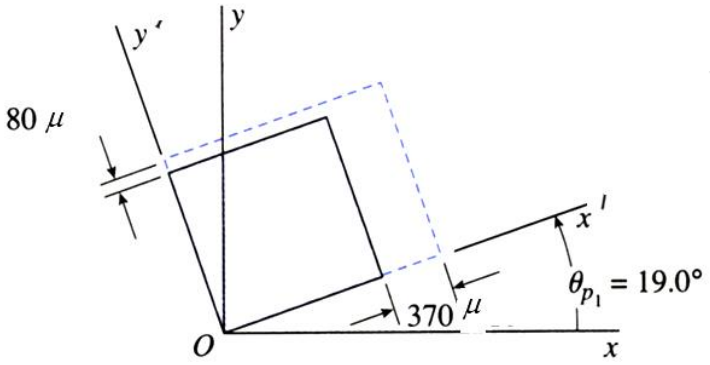
主應變 $\varepsilon_{1,2}=\dfrac{\varepsilon_x+\varepsilon_y}{2}\pm\sqrt{(\dfrac{\varepsilon_x-\varepsilon_y}{2})^2+(\dfrac{\gamma_{xy}}{2})^2}\Rightarrow \varepsilon_1=370\mu,\varepsilon_2=80\mu$

主軸方向的角度：$\tan(2\theta_p)=\dfrac{\gamma_{xy}}{\varepsilon_x-\varepsilon_y}=\dfrac{180}{340-110}=0.7826$

$\theta_p=19^\circ$及109°

將 $\theta_p=19^\circ$帶入 $\varepsilon_\theta=\dfrac{\varepsilon_x+\varepsilon_y}{2}+\dfrac{\varepsilon_x-\varepsilon_y}{2}\cos 2\theta+\dfrac{\gamma_{xy}}{2}\sin 2\theta\Rightarrow \varepsilon_1=370\mu$

將 $\theta_p=109^\circ$帶入 $\varepsilon_\theta=\dfrac{\varepsilon_x+\varepsilon_y}{2}+\dfrac{\varepsilon_x-\varepsilon_y}{2}\cos 2\theta+\dfrac{\gamma_{xy}}{2}\sin 2\theta\Rightarrow \varepsilon_2=80\mu$

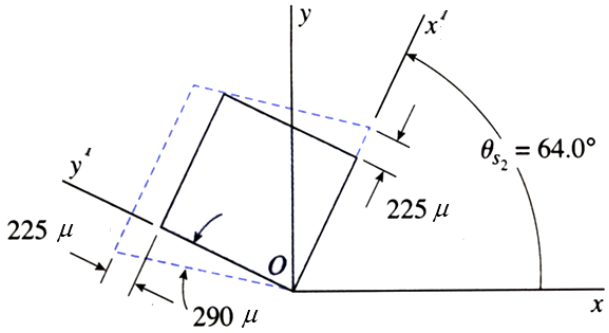


(c)

最大剪應變 $\gamma_{\max}=\varepsilon_1-\varepsilon_2=290\mu$

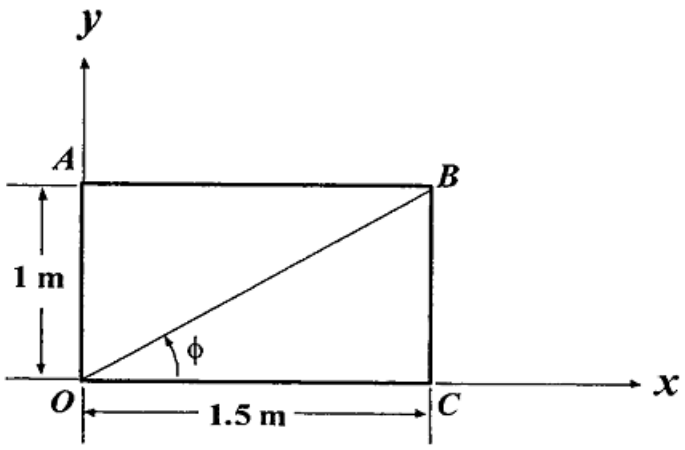
$\theta_{s1}=\theta_p+45^\circ=64^\circ$帶入 $\dfrac{\gamma_\theta}{2}=-\dfrac{\varepsilon_x-\varepsilon_y}{2}\sin 2\theta+\dfrac{\gamma_{xy}}{2}\cos 2\theta\Rightarrow \gamma_{\min}=-290\mu$

$\theta_{s2}=\theta_{s1}-45^\circ=-26^\circ$帶入 $\dfrac{\gamma_\theta}{2}=-\dfrac{\varepsilon_x-\varepsilon_y}{2}\sin 2\theta+\dfrac{\gamma_{xy}}{2}\cos 2\theta\Rightarrow \gamma_{\max}=290\mu$


y
x′
y′
θs2 = 64.0°
225 μ
225 μ
O
290 μ
x

範例 4

A rectangular plate $OABC$ as shown below is known to have uniform strain components: $\varepsilon_x = 520\times10^{-6}$, $\varepsilon_y = 250\times10^{-6}$ and $\gamma_{xy} = 400\times10^{-6}$. Determine (a) the principal strains and the corresponding directions, (b) the change in the length of diagonal OB and



【103 清大動機】

【解析】

(a)

$$\varepsilon_{1,2} = \frac{\varepsilon_x + \varepsilon_y}{2} \pm \sqrt{(\frac{\varepsilon_x - \varepsilon_y}{2})^2 + (\frac{\gamma_{xy}}{2})^2}$$

$$\Rightarrow \varepsilon_{1,2} = \frac{520+250}{2} \pm \sqrt{(\frac{520-250}{2})^2 + (\frac{400}{2})^2} = 626.3\mu, 143.7\mu$$

$$\tan(2\theta_p) = \frac{\gamma_{xy}}{\varepsilon_x - \varepsilon_y} = \tan(2\theta_p) = \frac{400}{500-250} \Rightarrow \theta_{p1} = 28^\circ$$

$$\theta_{p2} = -62^\circ$$

(b)

$$\varepsilon_\theta = \varepsilon_b = \frac{\varepsilon_x + \varepsilon_y}{2} + \frac{\varepsilon_x - \varepsilon_y}{2}\cos 2\theta + \frac{\gamma_{xy}}{2}\sin 2\theta = \Delta d$$

$$\Rightarrow \frac{520+(250)}{2} + \frac{520-(250)}{2}\cos(33.7^\circ) + \frac{\gamma_{xy}}{2}\sin(2\times33.7^\circ) = 621.52\times10^{-6}$$

$$\Delta L = 621.52\times10^{-6}\times\sqrt{1000^2+1500^2} = 1.12(mm)$$

範例 5

The strains for an element of material in plane strain are as follows: $\varepsilon_x = 480\mu$, $\varepsilon_y = 140\mu$, and $\gamma_{xy} = -350\mu$. Determine the principal strains and maximum shearstrains, and show these strains on sketches of properly oriented elements.

【台大工科】

【中文翻譯】

構件內某一點所受之平面應變如下：$\varepsilon_x = 480\mu$，$\varepsilon_y = 140\mu$，$\gamma_{xy} = -350\mu$。試求(a)主應變，(b)同平面最大剪應變。將上列結果以應變元素將變形前後之圖形繪出。

【解析】

標上參考點 其具座標 $(\varepsilon_x = 480\mu \ , \ \frac{\gamma_{xy}}{2} = -175\mu)$，連接點及圓心，此距離為此圓之半徑

$$R = \sqrt{(\frac{\varepsilon_x - \varepsilon_y}{2})^2 + (\frac{\gamma_{xy}}{2})^2} = 244\mu$$

$$\varepsilon_{av} = \frac{\varepsilon_x + \varepsilon_y}{2} = 310\mu$$

$$\varepsilon_1 = \frac{\varepsilon_x + \varepsilon_y}{2} + R = 554\mu \qquad \varepsilon_2 = \frac{\varepsilon_x + \varepsilon_y}{2} - R = 66\mu$$

$$\tan 2\theta_p = \frac{\gamma_{xy}}{\varepsilon_x - \varepsilon_y} = -1.0294$$

$$2\theta_p = -45.8° \quad \text{and} \quad 134.2°$$

$$\theta_p = -22.9° \quad \text{and} \quad 67.1°$$

$$\theta_{p_1} = -22.9° \qquad \varepsilon_1 = 554 \times 10^{-6}$$

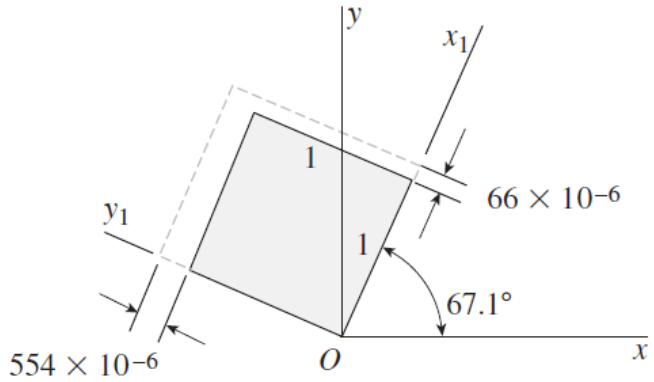
$$\theta_{p_2} = 67.1° \qquad \varepsilon_2 = 66 \times 10^{-6}$$

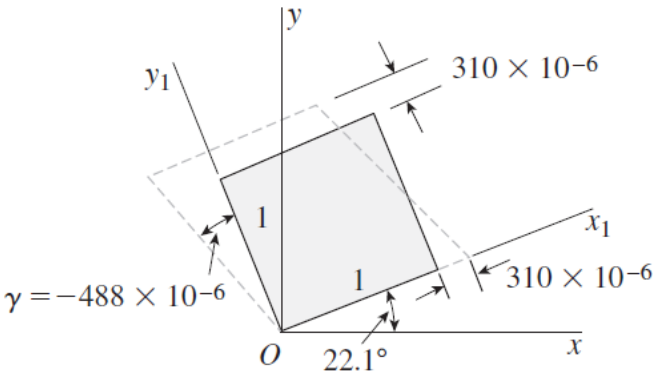
$$\gamma_{\max} = 2R = 2 \times \sqrt{(\frac{\varepsilon_x - \varepsilon_y}{2})^2 + (\frac{\gamma_{xy}}{2})^2} = 488\mu$$

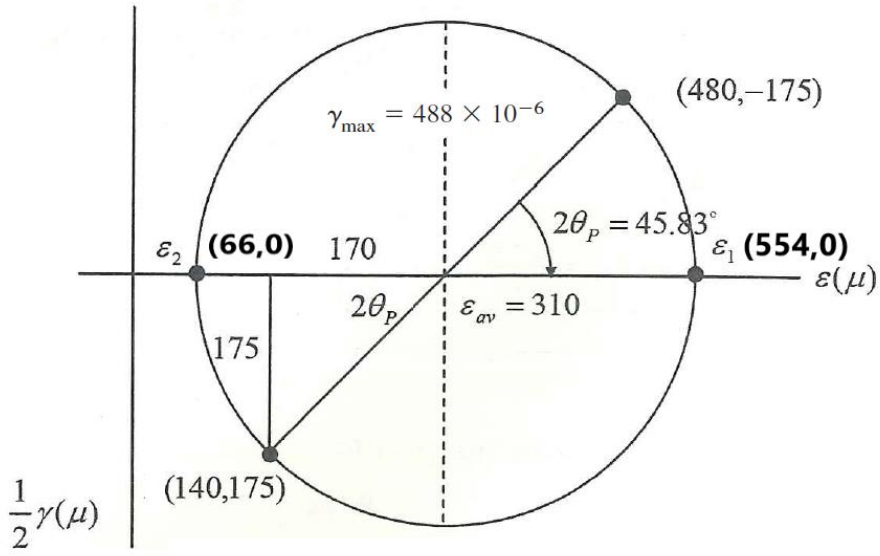
$$\theta_{s_1} = \theta_{p_1} - 45° = -67.9° \text{ or } 112.1°$$

$$\theta_{s_2} = \theta_{s_1} + 90° = 22.1°$$

$$\gamma_{\min} = -488 \times 10^{-6}$$

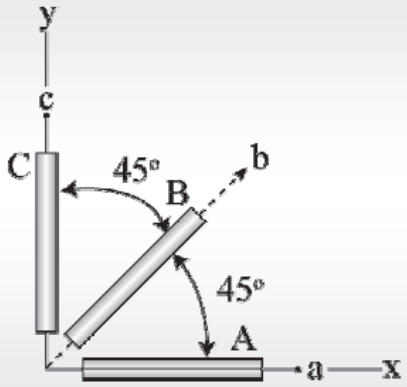

y
x_1
1
y_1
1
66×10^{-6}
67.1°
554×10^{-6}
O
x


y
y_1
310×10^{-6}
1
x_1
1
310×10^{-6}
$\gamma = -488 \times 10^{-6}$
O
22.1°
x


$\gamma_{max} = 488 \times 10^{-6}$
(480,–175)
$2\theta_P = 45.83°$
ε_2
(66,0)
170
ε_1 (554,0)
$\varepsilon(\mu)$
$2\theta_P$
$\varepsilon_{av} = 310$
175
(140,175)
$\frac{1}{2}\gamma(\mu)$

範例 6

一菊花型應變規含三枚電阻式應變規，安排如右圖所示。A、B 及 C 規對 O_a、O_b 及 O_c 軸所測得之正向應變分別為 ε_a、ε_b 及 ε_c，試求對 xy 軸的應變 ε_x、ε_y 及 γ_{xy}。(96 年機械高考)



【105 北科】【106 成大工科】

【解析】

解 $\varepsilon_a = \varepsilon_x$, $\varepsilon_c = \varepsilon_y$

$\varepsilon_b = \varepsilon_{45°}$

$$= \frac{\varepsilon_x + \varepsilon_y}{2} + \frac{\varepsilon_x - \varepsilon_y}{2}\cos(2\times 45°) + (\frac{r_{xy}}{2})\sin(2\times 45°)$$

$$\Rightarrow \varepsilon_b = \frac{\varepsilon_a + \varepsilon_c}{2} + (\frac{r_{xy}}{2})$$

$$\Rightarrow r_{xy} = 2\varepsilon_b - \varepsilon_a - \varepsilon_c$$

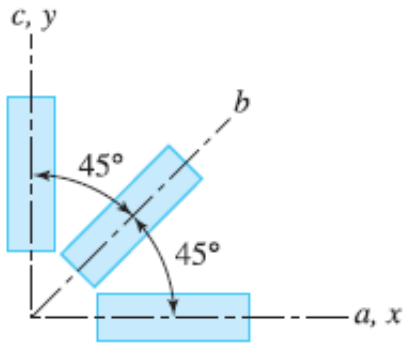
範例 7

圖中應變計 A、B、C 量得 O 點在 A、B、C 三方向之正向應變（normal strain）為：

$\varepsilon_a = 500\times10^{-6}$，$\varepsilon_b = 360\times10^{-6}$，$\varepsilon_c = -70\times10^{-6}$

㈠求 O 點之主應變，並繪主應變下之元素變形圖。（15 分）

㈡求 O 點之最大剪應變 γ_{max}，並繪最大剪應變下之元素變形圖。（10 分）



【土木四等考試】

【解析】

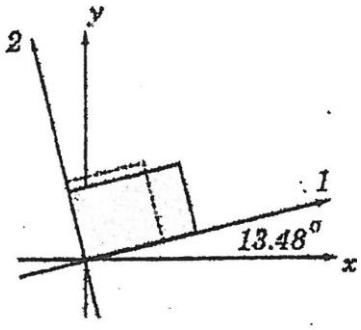
$\varepsilon_y = -70\mu = \varepsilon_c$ 、 $\varepsilon_x = 500\mu = \varepsilon_a$

$$\varepsilon_\theta = \varepsilon_b = \frac{\varepsilon_x+\varepsilon_y}{2} + \frac{\varepsilon_x-\varepsilon_y}{2}\cos 2\theta + \frac{\gamma_{xy}}{2}\sin 2\theta = 360\mu$$

$$\Rightarrow \frac{500-70}{2} + \frac{500-(-70)}{2}\cos(2\times45^\circ) + \frac{\gamma_{xy}}{2}\sin(2\times45^\circ) = 360\mu \Rightarrow \gamma_{xy} = 290\mu$$

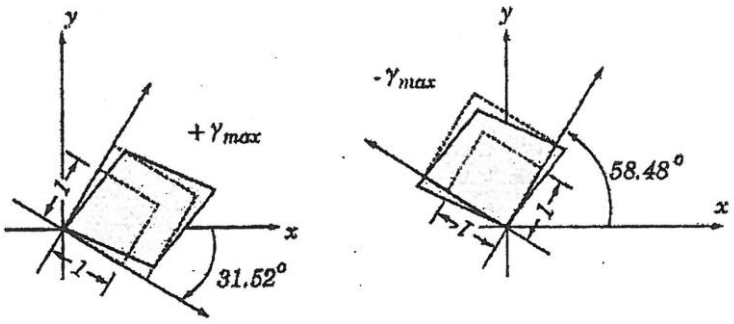
主應變 $\varepsilon_{1,2} = \frac{\varepsilon_x+\varepsilon_y}{2} \pm \sqrt{(\frac{\varepsilon_x-\varepsilon_y}{2})^2 + (\frac{\gamma_{xy}}{2})^2} \Rightarrow \varepsilon_1 = 534.76\mu, \varepsilon_2 = -104.76\mu$

$$\tan(2\theta_p) = \frac{\gamma_{xy}}{\varepsilon_x-\varepsilon_y} \Rightarrow \theta_{p1} = 13.48^\circ \text{ 、 } \theta_{p2} = 103.48^\circ$$



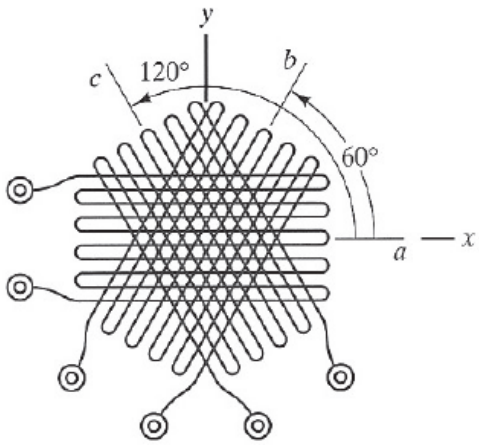
最大剪應變 $\gamma_{max} = \varepsilon_1 - \varepsilon_2 = 639.52\mu$

$$\tan(2\theta_s) = -\frac{\varepsilon_y - \varepsilon_x}{\gamma_{xy}} \Rightarrow \theta_{s1} = -31.52^\circ \text{ 、 } \theta_{s2} = 58.48^\circ$$



範例 8

右圖所示三軸花瓣應變計所量取之應變分別為 $\varepsilon_a = 0.004, \varepsilon_b = -0.003, \varepsilon_c = 0.002$，則 ε_y 等於：



(A) -0.001

(B) -0.002

(C) -0.0058

(D) -0.007。

(93 年機械高考第一試)

【台科機械】

【解析】

3. (B)。 $\varepsilon_x = \varepsilon_a = 0.004$

$$\varepsilon_b = \varepsilon_{60°} = \frac{\varepsilon_x + \varepsilon_y}{2} + \frac{\varepsilon_x - \varepsilon_y}{2}\cos(2\times 60°) + (\frac{r_{xy}}{2})\sin(2\times 60°)$$

$$\Rightarrow -0.003 = \frac{0.004 + \varepsilon_y}{2} + \frac{0.004 - \varepsilon_y}{2}\cos(120°) + (\frac{r_{xy}}{2})\sin 120°$$

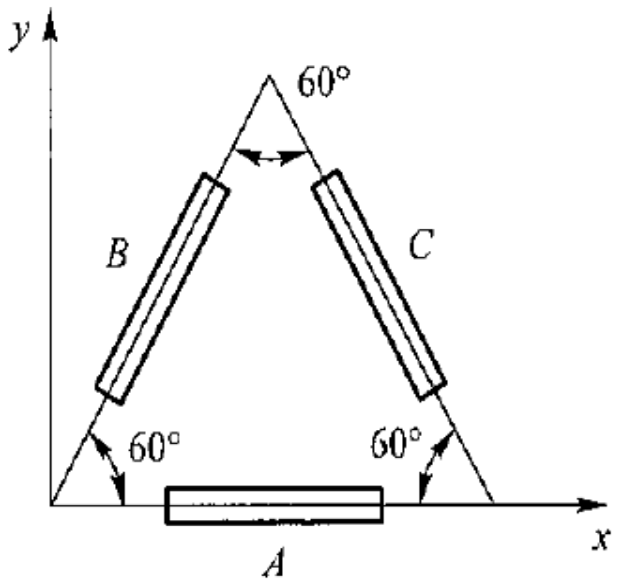
$$\varepsilon_c = \varepsilon_{120°} = \frac{\varepsilon_x + \varepsilon_y}{2} + \frac{\varepsilon_x - \varepsilon_y}{2}\cos(2\times 120°) + (\frac{r_{xy}}{2})\sin(2\times 120°)$$

$$\Rightarrow 0.002 = \frac{0.004 + \varepsilon_y}{2} + \frac{0.004 - \varepsilon_y}{2}\cos(240°) + (\frac{r_{xy}}{2})\sin(240°)$$

求得 $\varepsilon_y = -0.002 \quad r_{xy} = -5.77\times 10^{-3}$

範例 9

A 60° strain rosette, or delta rosette, consists of three electrical-resistance strain gages arranged as shown in the figure. Gage A measures the normal strain ε_a in the direction of the x axis. Gages B and C measure the strains ε_b and ε_c in the inclined directions shown. Obtain the equations for the strains ε_x, ε_y, and γ_{xy} associated with the xy axes.



【機械高考】

【中文翻譯】

60°的三角形應變簇組由圖示的電阻式應變計組成。A量測x軸方向上正向應變ε_{xx}，B及C測量傾斜方向上的應變ε_b及ε_c。試求平面應變分量。

【解析】

$\theta = 0°$ Strain $= \varepsilon_A$

$\theta = 60°$ Strain $= \varepsilon_B$

$\theta = 120°$ Strain $= \varepsilon_C$

$$\varepsilon_{x_1} = \frac{\varepsilon_x + \varepsilon_y}{2} + \frac{\varepsilon_x - \varepsilon_y}{2}\cos 2\theta + \frac{\gamma_{xy}}{2}\sin 2\theta$$

$$\varepsilon_B = \frac{\varepsilon_A + \varepsilon_y}{2} + \frac{\varepsilon_A - \varepsilon_y}{2}(\cos 120°) + \frac{\gamma_{xy}}{2}(\sin 120°)$$

$$\varepsilon_B = \frac{\varepsilon_A}{4} + \frac{3\varepsilon_y}{4} + \frac{\gamma_{xy}\sqrt{3}}{4}$$

$$\varepsilon_{x_1} = \frac{\varepsilon_x + \varepsilon_y}{2} + \frac{\varepsilon_x - \varepsilon_y}{2}\cos 2\theta + \frac{\gamma_{xy}}{2}\sin 2\theta$$

$$\varepsilon_C = \frac{\varepsilon_A + \varepsilon_y}{2} + \frac{\varepsilon_A - \varepsilon_y}{2}(\cos 240°) + \frac{\gamma_{xy}}{2}(\sin 240°)$$

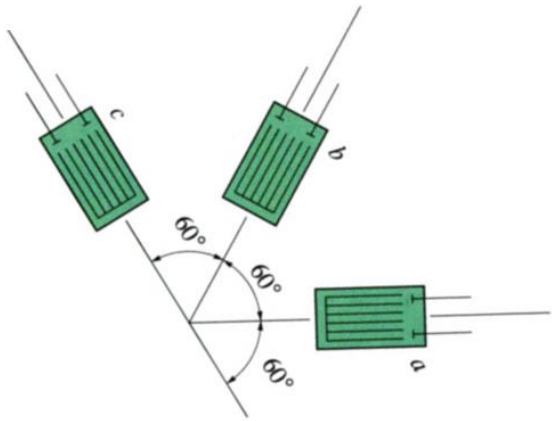
$$\varepsilon_C = \frac{\varepsilon_A}{4} + \frac{3\varepsilon_y}{4} - \frac{\gamma_{xy}\sqrt{3}}{4}$$

$$\varepsilon_y = \frac{1}{3}(2\varepsilon_B + 2\varepsilon_C - \varepsilon_A)$$

$$\gamma_{xy} = \frac{2}{\sqrt{3}}(\varepsilon_B - \varepsilon_C)$$

範例 10

The 60° strain rosette is mounted on the bracket. The following readings are obtained for each gauge: $\varepsilon_a = -100\mu$, $\varepsilon_b = 250\mu$, and $\varepsilon_c = 150\mu$ and Determine (a) the principal strains and (b) the maximum inplane shear strain and associated average normal strain.



【交大機械】

【解析】

$$\varepsilon_x = \varepsilon_a = -100(10^{-6})$$

$$\varepsilon_y = \frac{1}{3}\left(2\varepsilon_b + 2\varepsilon_c - \varepsilon_a\right)$$

$$= \frac{1}{3}\left[2(250) + 2(150) - (-100)\right](10^{-6})$$

$$= 300(10^{-6})$$

$$\gamma_{xy} = \frac{2}{\sqrt{3}}(\varepsilon_b - \varepsilon_c) = \frac{2}{\sqrt{3}}(250 - 150)(10^{-6}) = 115.47(10^{-6})$$

$$\varepsilon_{\text{avg}} = \frac{\varepsilon_x + \varepsilon_y}{2} = \left(\frac{-100 + 300}{2}\right)(10^{-6}) = 100(10^{-6})$$

$$\varepsilon_{1,2} = \frac{\varepsilon_x + \varepsilon_y}{2} \pm \sqrt{\left(\frac{\varepsilon_x - \varepsilon_y}{2}\right)^2 + \left(\frac{\gamma_{xy}}{2}\right)^2}$$

$$\Rightarrow \varepsilon_{1,2} = \frac{-100+300}{2} \pm \sqrt{(\frac{-100-300}{2})^2 + (\frac{115.47}{2})^2} = 308\mu, -108\mu$$

$$\tan(2\theta_p) = \frac{\gamma_{xy}}{\varepsilon_x - \varepsilon_y} = \tan(2\theta_p) = \frac{115.47}{-100-300} \Rightarrow \theta_{p1} = -8.05^\circ$$

3-3. 應變張量(strain tensor)

一、應變位移場(displacement field)

材料體變形時，各材料點同時也發生位移，每個材料點的位移通常並不相同，因此在材料體內就形成一個位移分佈稱之為位移場(displacement field)，如圖所示，若以向量 r 來表示某材料點的位置，而u表示此材料點的位移向量，即 u 為r的函數$u=u(r)$也稱之為位移場函數(displacement field function)

對應於材料體內不同的位置，通常有不同的位移向量，其中位置向量 r 及位移向量 u 如下: $r=x\vec{i}+y\vec{j}+z\vec{k}$ 、 $u(r)=u_x\vec{i}+u_y\vec{j}+u_z\vec{k}$ ，其中u_x u_y u_z 亦為 x、y、z 的座標即:

$u_x=u_x(\mathrm{x},\mathrm{y},\mathrm{z})$ 、 $u_y=u_y(\mathrm{x},\mathrm{y},\mathrm{z})$ 、 $u_z=u_z(\mathrm{x},\mathrm{y},\mathrm{z})$

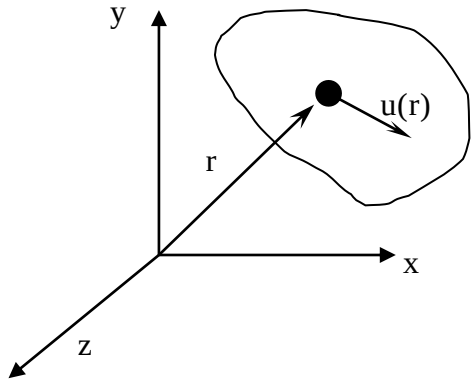
直角座標中位移場函數與應變函數關係如下:

$$\varepsilon_x=\frac{\partial u_x}{\partial x} \text{ 、 } \varepsilon_y=\frac{\partial u_y}{\partial y} \text{ 、 } \varepsilon_z=\frac{\partial u_z}{\partial z}$$

$$\gamma_{xy}=\gamma_{\mathrm{yx}}=\frac{\partial u_x}{\partial y}+\frac{\partial u_y}{\partial x}$$

$$\gamma_{xz}=\gamma_{\mathrm{zx}}=\frac{\partial u_x}{\partial z}+\frac{\partial u_z}{\partial x}$$

$$\gamma_{zy}=\gamma_{\mathrm{yz}}=\frac{\partial u_z}{\partial y}+\frac{\partial u_y}{\partial z}$$



位移場函數與應變函數關係證明如下:

如圖(a)所示材料元素投影在 xy 平面上為一$d_x\times \mathrm{d}_y$ 矩形區域，x 方向的位移量為$\frac{\partial u_x}{\partial x}d_x$、y 方向的位移量為$\frac{\partial u_y}{\partial y}d_y$，依照正向應變定義可得:

$$\varepsilon_x = \frac{\frac{\partial u_x}{\partial x} d_x}{d_x} = \frac{\partial u_x}{\partial x}$$

$$\varepsilon_y = \frac{\frac{\partial u_y}{\partial y} d_y}{d_y} = \frac{\partial u_y}{\partial y}$$

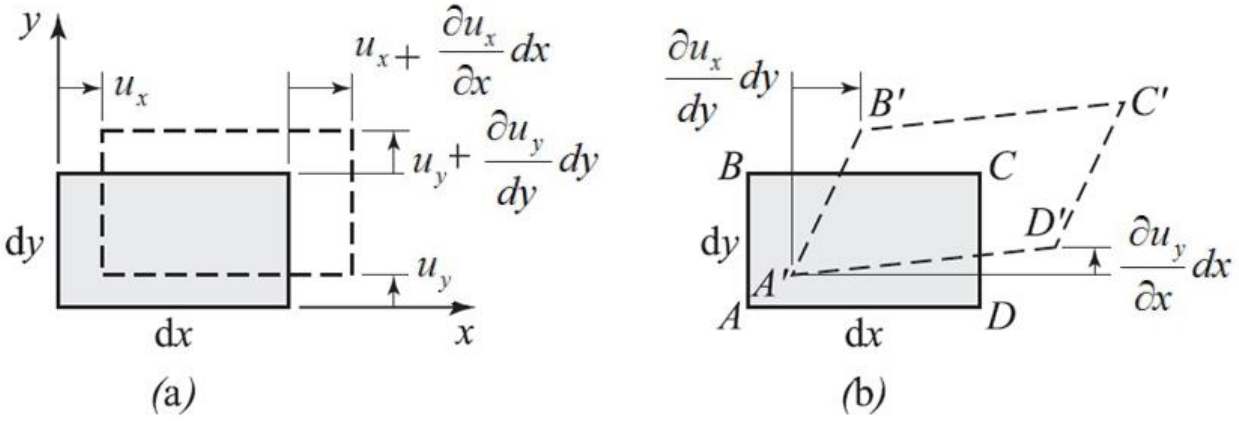
元素變形後矩形區域將會變成如圖(b)所示之形狀，線段 AB 變形為線段 A’ B’ ，其角度變化為 $\frac{\frac{\partial u_x}{\partial y} d_y}{d_y} = \frac{\partial u_x}{\partial y}$ ，同理線段 AD 變形為線段 A’ D’ ，其角度變化為 $\frac{\frac{\partial u_y}{\partial x} d_x}{d_x} = \frac{\partial u_y}{\partial x}$ ，故 A 點的剪應變為兩個角度和:

$$\gamma_{xy} = \gamma_{yx} = \frac{\partial u_x}{\partial y} + \frac{\partial u_y}{\partial x}$$

同理若考慮元素在 xz 平面上投影則可得:

$$\varepsilon_z = \frac{\frac{\partial u_z}{\partial z} d_z}{d_z} = \frac{\partial u_z}{\partial z}$$

$$\gamma_{zy} = \gamma_{yz} = \frac{\partial u_z}{\partial y} + \frac{\partial u_y}{\partial z}$$



二、應變張量(strain tensor)

由於剪應變在轉換時其正應變的係數多乘 2 倍，因此元素的應變 ε_x、ε_y、ε_z、γ_{yx}、γ_{xy} 、γ_{yz}、γ_{yz} 、γ_{xz}、γ_{zx} 沒有形成一個張量，所以只要把剪應變除以 2 則可以和 ε_x、ε_y 、ε_z 形成一個張量，我們定義應變張量 ε_{ij} 為:

$$\varepsilon_{xx}=\frac{\partial u_x}{\partial x}=\varepsilon_x \text{ 、 } \varepsilon_{yy}=\frac{\partial u_y}{\partial y}=\varepsilon_y \text{ 、 } \varepsilon_{zz}=\frac{\partial u_z}{\partial z}=\varepsilon_z$$

$$\varepsilon_{xy}=\frac{1}{2}\times(\frac{\partial u_x}{\partial y}+\frac{\partial u_y}{\partial x})=\frac{\gamma_{xy}}{2}$$

$$\varepsilon_{xz}=\frac{1}{2}\times(\frac{\partial u_x}{\partial z}+\frac{\partial u_z}{\partial x})=\frac{\gamma_{xz}}{2}$$

$$\varepsilon_{yz}=\frac{1}{2}\times(\frac{\partial u_y}{\partial z}+\frac{\partial u_z}{\partial y})=\frac{\gamma_{yz}}{2}$$

我們以x_1、x_2、x_3分別代替x、y、z，以u_1、u_2、u_3分別代替u_x、u_y、u_z，這樣的應變張量可以寫成以下通式

$$\varepsilon_{ij}=\frac{1}{2}\times(\frac{\partial u_i}{\partial x_j}+\frac{\partial u_j}{\partial x_i})\text{其中(i, j}=1,2,3)$$

上式若是ε_{ij} 下標相同的話表示正向應變如

$$\varepsilon_{11}=\frac{1}{2}\times(\frac{\partial u_1}{\partial x_1}+\frac{\partial u_1}{\partial x_1})=\frac{\partial u_1}{\partial x_1}=\frac{\partial u_x}{\partial x}=\varepsilon_x$$

則應變張量可寫成:

$$\varepsilon_{ij}=\begin{bmatrix}\varepsilon_{11} & \varepsilon_{12} & \varepsilon_{13}\\ \varepsilon_{21} & \varepsilon_{22} & \varepsilon_{23}\\ \varepsilon_{31} & \varepsilon_{32} & \varepsilon_{33}\end{bmatrix}=\begin{bmatrix}\varepsilon_{xx} & \varepsilon_{xy} & \varepsilon_{xz}\\ \varepsilon_{yx} & \varepsilon_{yy} & \varepsilon_{yz}\\ \varepsilon_{zx} & \varepsilon_{zy} & \varepsilon_{zz}\end{bmatrix}=\begin{bmatrix}\varepsilon_{xx} & \frac{\gamma_{xy}}{2} & \frac{\gamma_{xz}}{2}\\ \frac{\gamma_{yx}}{2} & \varepsilon_{yy} & \frac{\gamma_{yz}}{2}\\ \frac{\gamma_{zx}}{2} & \frac{\gamma_{zy}}{2} & \varepsilon_{zz}\end{bmatrix}$$

三、 應變諧和條件(相容方程式)

假設變形物體的應變函數ε_x、ε_y、γ_{yx}均有 2 階導數存在且連續，則此應變諧和條件(相容方程式)必滿足:

$$\frac{\partial^2 u_x}{\partial y^2}+\frac{\partial^2 u_y}{\partial x^2}=\frac{\partial^2 \gamma_{xy}}{\partial x\partial y}$$

證明:

$$\varepsilon_x=\frac{\partial u_x}{\partial x} \text{ 、 } \varepsilon_y=\frac{\partial u_y}{\partial y} \text{ 、 } \gamma_{xy}=\frac{\partial u_x}{\partial y}+\frac{\partial u_y}{\partial x}$$

假設存在且連續則應變的 2 階偏微分

$$\frac{\partial^2 \varepsilon_x}{\partial y^2}=\frac{\partial^2}{\partial y^2}(\frac{\partial u_x}{\partial x})=\frac{\partial^3 u_x}{\partial x\partial y^2}$$

$$\frac{\partial^2 \varepsilon_y}{\partial x^2}=\frac{\partial^2}{\partial x^2}(\frac{\partial u_y}{\partial y})=\frac{\partial^3 u_y}{\partial y\partial x^2}$$

$$\frac{\partial^2 \gamma_{xy}}{\partial x \partial y} = \frac{\partial^2}{\partial x \partial y}\left(\frac{\partial u_x}{\partial y} + \frac{\partial u_y}{\partial x}\right) = \frac{\partial^3 u_x}{\partial x \partial y^2} + \frac{\partial^3 u_y}{\partial y \partial x^2}$$

$$\Rightarrow \frac{\partial^2 u_x}{\partial y^2} + \frac{\partial^2 u_y}{\partial x^2} = \frac{\partial^2 \gamma_{xy}}{\partial x \partial y}$$

故得證

四、 三維應變轉換

應變矩陣中的元素值(亦即正向應變及剪應變)，與所採用的座標系有關，採用不同之座標系統時，應變值會隨之變化，由座標系方向餘弦所定義轉換矩陣$[A]$，當兩個座標系均為正交座標系時，其間的轉換矩陣$[A]$稱為正交矩陣，正交矩陣有一個很重要的特性為:

$[A]^T = [A]^{-1}$

應變向量可表示為

可得$[\varepsilon]' = [A][\varepsilon][A]^T$

五、 最大主應變

依據最大主應變的定義為在任一切面之應變分量，其面上沒有剪應變，僅有正向應變即為最大主應變，該切面為最大主應變面， 其中最大主應力ε即為特徵值:

$([\varepsilon] - \varepsilon[I]) = 0$(特徵方程式)

可寫為

$$\left|\begin{bmatrix} \varepsilon_{xx} - \varepsilon & \frac{\gamma_{xy}}{2} & \frac{\gamma_{xz}}{2} \\ \frac{\gamma_{yx}}{2} & \varepsilon_{yy} - \varepsilon & \frac{\gamma_{yz}}{2} \\ \frac{\gamma_{zx}}{2} & \frac{\gamma_{zy}}{2} & \varepsilon_{zz} - \varepsilon \end{bmatrix}\right| = 0$$

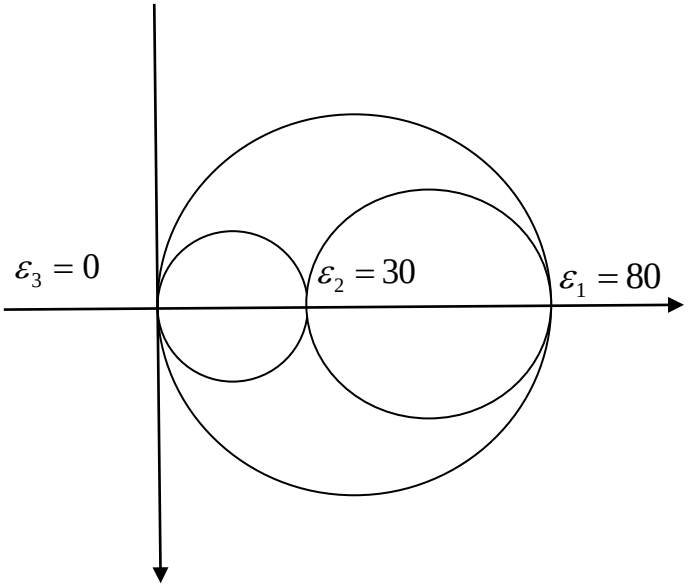
應變張量$[\varepsilon]$的特徵值可求出主應變ε_1、ε_2、ε_3

六、 3D 應變之莫爾圓

所謂三軸向應變就是 3D 應變元素在三個軸向只有正交應變存在，而無剪應變，如圖所示，若要求 xy 平面內的面內最大剪應變，就是將應變元素轉角度，方法如上面莫爾圓法所提，然而要是求絕對最大剪應變:

情況一: $\varepsilon_1 > \varepsilon_2 > \varepsilon_3$

平面應變經計算$\varepsilon_1 = 80$ 、 $\varepsilon_2 = 30$ ， $\varepsilon_z = \varepsilon_3 = 0$ ，如下圖所示

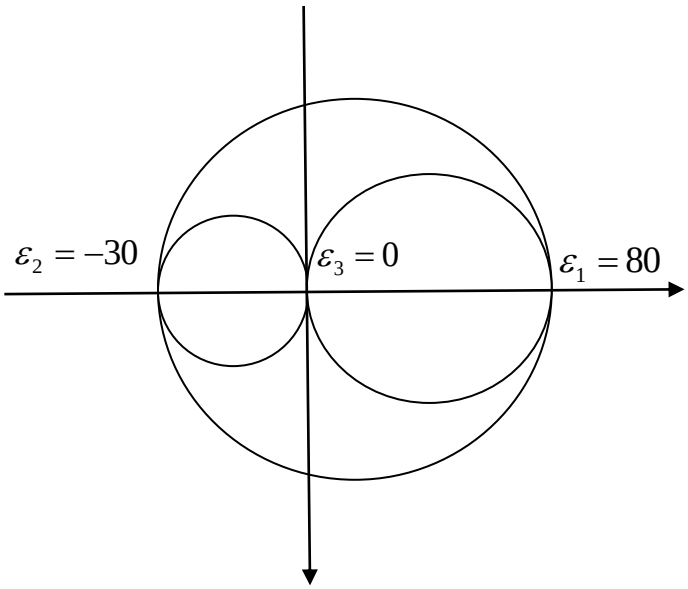


xy 平面最大剪應變 $\gamma_{\max}^{in-plane} = \varepsilon_1 - \varepsilon_2 = 50$

絕對最大剪應變 $\gamma_{\max}^{abs} = \varepsilon_1 - \varepsilon_3 = 80$

情況二： $\varepsilon_1 > \varepsilon_3 > \varepsilon_2$

平面應變經計算 $\varepsilon_1 = 80$ 、 $\varepsilon_2 = -30$ ， $\varepsilon_3 = 0$ ，如下圖所示

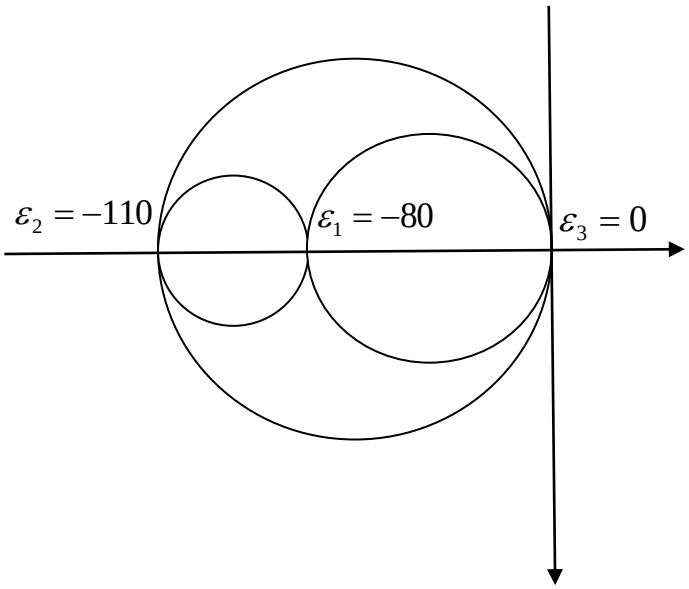


xy 平面最大剪應變 $\gamma_{\max}^{in-plane} = \varepsilon_1 - \varepsilon_2 = 110$

絕對最大剪應變 $\gamma_{\max}^{abs} = \varepsilon_1 - \varepsilon_2 = 110$

情況三： $\varepsilon_3 > \varepsilon_1 > \varepsilon_2$

平面應變經計算 $\varepsilon_1 = -80$ 、 $\varepsilon_2 = -110$ ， $\varepsilon_3 = 0$ ，如下圖所示



xy 平面最大剪應變 $\gamma_{\max}^{in-plane} = \varepsilon_1 - \varepsilon_2 = 30$

絕對最大剪應變 $\gamma_{\max}^{abs} = \varepsilon_3 - \varepsilon_2 = 110$

範例 11

The following two dimensional displacement field describes the movement of a body under load :

$$u_x = x^2 + y^2, \quad u_y = 3 + xy \quad (\times 10^{-2}\ \text{ft})$$

Compute the normal strain at (0,1) in a direction $\vec{s}$ given as $\vec{s} = 0.60\vec{i} + 0.80\vec{j}$

【成大機械】

【中文翻譯】

下列二維的位移場描述一物體受力後的移動

$$u_x = x^2 + y^2 \quad ; \quad u_y = 3 + xy \qquad (\text{單位：}\ 10^{-2}\ ft)$$

試計算在點 $P(0,1)\ ft$ 處，沿 $\boldsymbol{e} = 0.6\boldsymbol{i} + 0.8\boldsymbol{j}$ 方向的正向應變。

【解析】

$$\varepsilon_x = \frac{\partial u_x}{\partial x} = (2x)\times 10^{-2} \quad ; \quad \varepsilon_y = \frac{\partial u_y}{\partial y} = (x)\times 10^{-2}$$

$$\gamma_{xy} = \gamma_{yx} = \frac{\partial u_x}{\partial y} + \frac{\partial u_y}{\partial x} = (2y + y)\times 10^{-2}$$

$x = 0$ ， $y = 1$

$$\frac{\partial u_x}{\partial x} = 2x = 0$$

$$\frac{\partial u_y}{\partial y} = x = 0$$

$$\frac{1}{2}\left(\frac{\partial u_x}{\partial y} + \frac{\partial u_y}{\partial x}\right) = \frac{1}{2}(2y + y)\times 10^{-2} = 1.5\times 10^{-2}$$

單位向量 $\boldsymbol{e}$ 與 x 軸的夾角 θ 爲

$$\theta = tan^{-1}\left(\frac{0.8}{0.6}\right) = 53.13^o$$

$$\varepsilon = \frac{\varepsilon_x + \varepsilon_y}{2} + \frac{\varepsilon_x - \varepsilon_y}{2}cos\,2\theta + \frac{\gamma_{xy}}{2}sin\,2\theta$$

$$= \left(1.5\times 10^{-2}\right)sin\,106.26^o = 1.44\times 10^{-2}$$

另解:

$$\begin{pmatrix}0.60 & 0.80\end{pmatrix}\begin{bmatrix}0 & 1.5\times10^{-2}\\ 1.5\times10^{-2} & 0\end{bmatrix}\begin{pmatrix}0.60\\ 0.80\end{pmatrix}=1.44\times10^{-2}$$

實戰演練

例 1

The strains for an element of material in plane strain are as follows: $\varepsilon_x = 480\mu$, $\varepsilon_y = 70\mu$, and $\gamma_{xy} = 420\mu$ .Determine (a)the principal strains and (b) maximum shearstrains, (c) the strains for an element oriented at an angle θ=75° and show these strains on sketches of properly oriented elements.

【中文翻譯】

構件內某一點所受之平面應變如下：ε_x = 480μ，$\varepsilon_y = 70\mu$ ，$\gamma_{xy} = 420\mu$ 。試求(a)主應變，(b)同平面最大剪應變，(c)θ = 75°方位之應變狀態。將上列結果以應變元素將變形前後之圖形繪出

【解析】

(a)主應變：

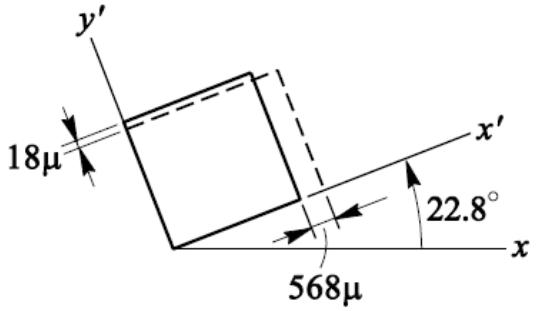
$$\tan 2\theta_p = \frac{\gamma_{xy}}{\varepsilon_x - \varepsilon_y} = \frac{420}{480-70} = \frac{420}{410}$$

$2\theta_{p1} = 45.7°$ ($\sin 2\theta_{p1} > 0$ ， $\cos 2\theta_{p1} > 0$)， $\theta_{p1} = 22.8°$ (逆時針方向)

$\varepsilon_x = 480\mu$， $\varepsilon_y = 70\mu$

主應變 $\varepsilon_{1,2} = \frac{\varepsilon_x + \varepsilon_y}{2} \pm \sqrt{(\frac{\varepsilon_x - \varepsilon_y}{2})^2 + (\frac{\gamma_{xy}}{2})^2} \Rightarrow \varepsilon_1 = 568\mu, \varepsilon_2 = -18\mu$

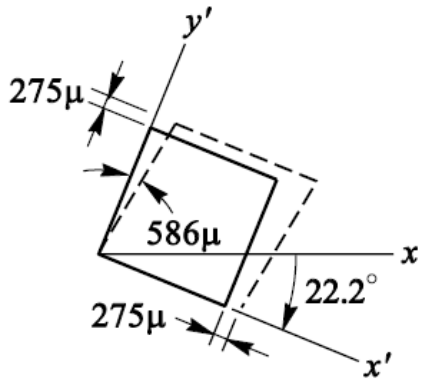
最大剪應變 $\gamma_{\max} = \varepsilon_1 - \varepsilon_2 = 586\mu$



(b)同平面最大剪應變

$\theta_{s1} = \theta_{p1} - 45° = 22.8° - 45° = -22.2°$ (順時針方向)

$$\varepsilon_{x'} = \varepsilon_{y'} = \varepsilon_{av} = \frac{\varepsilon_x + \varepsilon_y}{2} = 275\mu$$



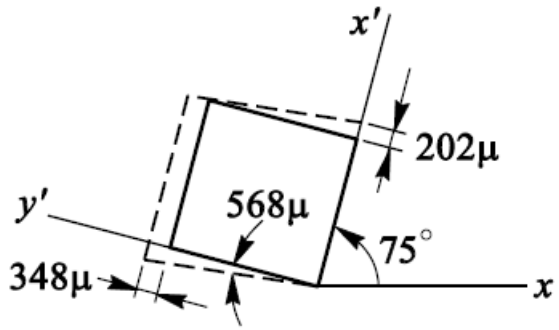
(c) $\theta = 75°$

$$\varepsilon_{x'} = \frac{\varepsilon_x + \varepsilon_y}{2} + \frac{\varepsilon_x - \varepsilon_y}{2}\cos 2\theta + \frac{\gamma_{xy}}{2}\sin 2\theta = 275\mu + 205\mu\cos 150° + 210\mu\sin 150° = 202\mu$$

$$\frac{\gamma_{x'y'}}{2} = -\frac{\varepsilon_x - \varepsilon_y}{2}\sin 2\theta + \frac{\gamma_{xy}}{2}\cos 2\theta = -205\mu\sin 150° + 210\mu\cos 150° = -284\ \mu$$

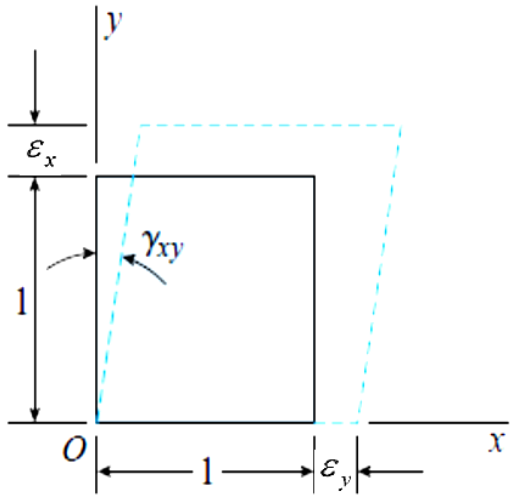
$$\gamma_{x'y'} = -568\ \mu$$

$$\varepsilon_{y'} = (\varepsilon_x + \varepsilon_y) - \varepsilon_{x'} = (480\mu + 70\mu) - 202\mu = 348\mu$$



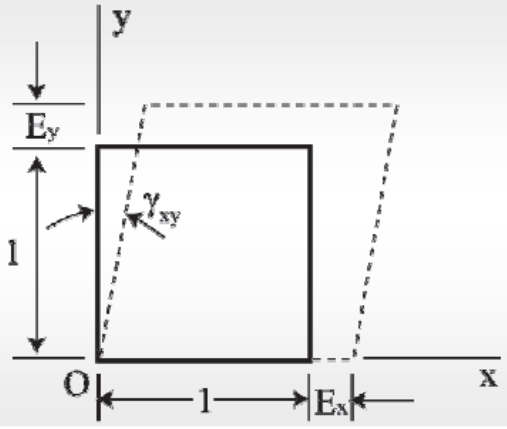
例 2

An element of material subjected to plane strain (see figure) has strains as follows: $\varepsilon_x = 220\mu$, $\varepsilon_y = 480\mu$, and $\gamma_{xy} = 180\mu$. Calculate the strains for an element oriented at an angle $\theta = 50^\circ$ and show these strains on a sketch of a properly oriented element.



【中文說明】

一材料元素受平面應變(見圖)有應變如下：$\varepsilon_x = 220\times10^{-6}$, $\varepsilon_y = 480\times10^{-6}$ ，及 $\gamma_{xy} = 180\times10^{-6}$ 。求指向 $\theta = 50°$ 角之元素的應變，並且在正確指向的元素上圖示這些應力。



【解析】

解 $\varepsilon_x = 220\times10^{-6}$, $\varepsilon_y = 480\times10^{-6}$, $\gamma_{xy} = 180\times10^{-6}$

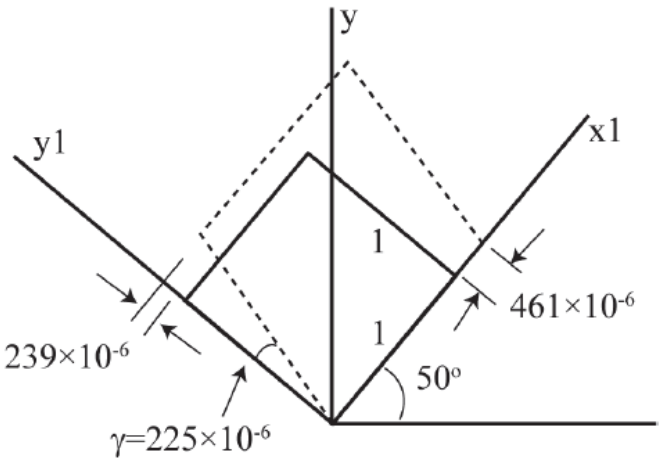
$$\varepsilon_{x_1} = (\varepsilon_x + \varepsilon_y)/2 + [(\varepsilon_x - \varepsilon_y)/2]\cos 2\theta + (\gamma_{xy}/2)\sin 2\theta$$

$$\gamma_{x_1y_1} = -[(\varepsilon_x - \varepsilon_y)/2]\cos 2\theta + (\gamma_{xy}/2)\sin 2\theta$$

$$\varepsilon_{y_1} = \varepsilon_x + \varepsilon_y + \varepsilon_z$$

for $\theta = 50°$: $\varepsilon_x = 461\times10^{-6}$, $\gamma_{x_1y_1} = 225\times10^{-6}$

$$\varepsilon_{y_1} = 239\times10^{-6}$$



例 3

The strains for an element of material in plane strain are as follows: $\varepsilon_x = 900\mu$, $\varepsilon_y = -333\mu$, and $\gamma_{xy} = 982\mu$ .Determine (a)the principal strains and (b) maximum shearstrains, and show these strains on sketches of properly oriented elements.

【中文翻譯】

構件內某一點所受之平面應變如下：$\varepsilon_x = 900\mu$，$\varepsilon_y = -333\mu$，$\gamma_{xy} = 982\mu$。試求(a)主應變，(b)同平面最大剪應變。將上列結果以應變元素將變形前後之圖形繪出。

【解析】

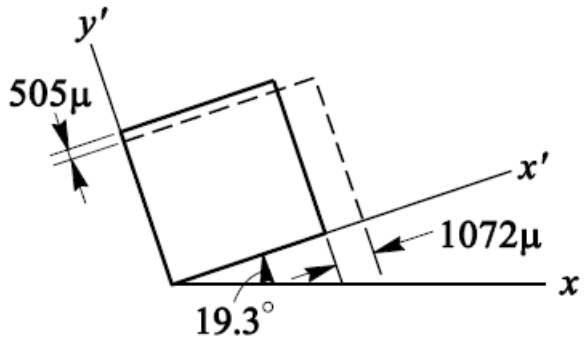
(a)主應變方向：$\tan 2\theta_p = \dfrac{\gamma_{xy}}{\varepsilon_x - \varepsilon_y} = \dfrac{982\mu}{900\mu - (-333\mu)} = \dfrac{982}{1233}$

$2\theta_{p1} = 38.53^\circ$ ($\sin 2\theta_{p1} > 0$ ， $\cos 2\theta_{p1} > 0$)， $\theta_{p1} = 19.3^\circ$ (逆時針方向)

主應變大小：$\varepsilon_{1,2} = \dfrac{\varepsilon_x + \varepsilon_y}{2} \pm \sqrt{\left(\dfrac{\varepsilon_x - \varepsilon_y}{2}\right)^2 + \left(\dfrac{\gamma_{xy}}{2}\right)^2}$

$= \dfrac{900\mu - 333\mu}{2} \pm \sqrt{\left(\dfrac{900\mu + 333\mu}{2}\right)^2 + \left(\dfrac{982\mu}{2}\right)^2} = 283.5\mu \pm 788.1\mu$

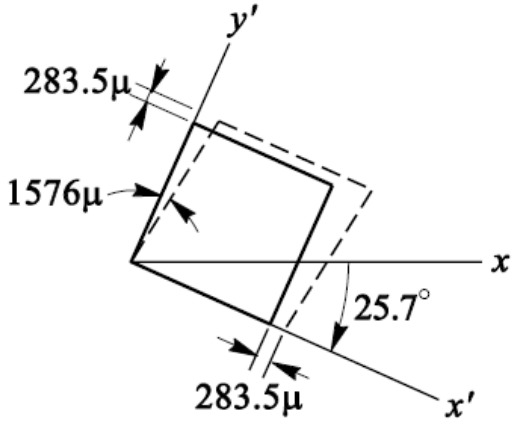
$\varepsilon_1 = 1072\mu$ ， $\varepsilon_2 = -505\mu$



(b)同平面最大剪應變：$\theta_{s1} = \theta_{p1} - 45^\circ = -25.7^\circ$ (順時針方向)

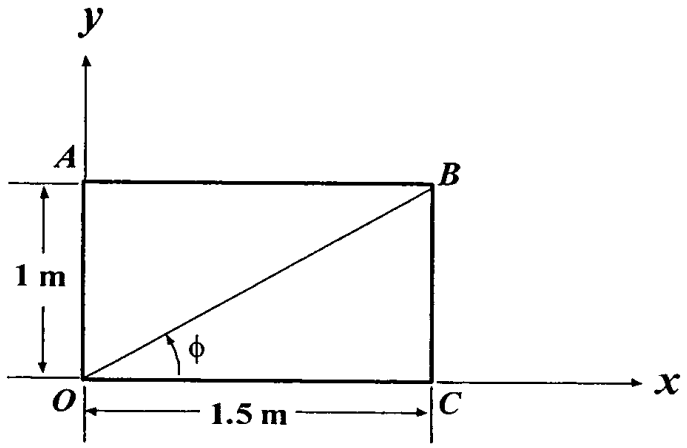
$\dfrac{\gamma_{\max}}{2} = \sqrt{\left(\dfrac{\varepsilon_x - \varepsilon_y}{2}\right)^2 + \left(\dfrac{\gamma_{xy}}{2}\right)^2} = 788.1\mu$ ， $\gamma_{\max} = 1576\mu$

$\varepsilon_{x'} = \varepsilon_{y'} = \dfrac{\varepsilon_x + \varepsilon_y}{2} = 283.5\mu$


y′
283.5μ
1576μ
x
25.7°
283.5μ
x′

例 4

A rectangular plate $OABC$ as shown below is known to have uniform strain components: $\varepsilon_x = 520\times10^{-6}$, $\varepsilon_y = 250\times10^{-6}$ and $\gamma_{xy} = 400\times10^{-6}$. Determine (a) the principal strains and the corresponding directions, (b) the change in the length of diagonal OB and (c) the change in the angle ϕ between diagonal OB and the x axis. (20%)



【103 清大動機】

【解析】

(a)

$$\varepsilon_{1,2} = \frac{\varepsilon_x + \varepsilon_y}{2} \pm \sqrt{(\frac{\varepsilon_x - \varepsilon_y}{2})^2 + (\frac{\gamma_{xy}}{2})^2}$$

$$\Rightarrow \varepsilon_{1,2} = \frac{520+250}{2} \pm \sqrt{(\frac{520-250}{2})^2 + (\frac{400}{2})^2} = 626.3\mu, 143.7\mu$$

$$\tan(2\theta_p) = \frac{\gamma_{xy}}{\varepsilon_x - \varepsilon_y} = \tan(2\theta_p) = \frac{400}{500-250} \Rightarrow \theta_{p1} = 28^\circ$$

$$\theta_{p2} = -62^\circ$$

(b)

$$\varepsilon_\theta = \varepsilon_b = \frac{\varepsilon_x + \varepsilon_y}{2} + \frac{\varepsilon_x - \varepsilon_y}{2}\cos 2\theta + \frac{\gamma_{xy}}{2}\sin 2\theta = \Delta d$$

$$\Rightarrow \frac{520+(250)}{2} + \frac{520-(250)}{2}\cos(33.7^\circ) + \frac{\gamma_{xy}}{2}\sin(2\times33.7^\circ) = 621.52\times10^{-6}$$

$$\Delta L = 621.52\times10^{-6}\times\sqrt{1000^2+1500^2} = 1.12(mm)$$

(c)

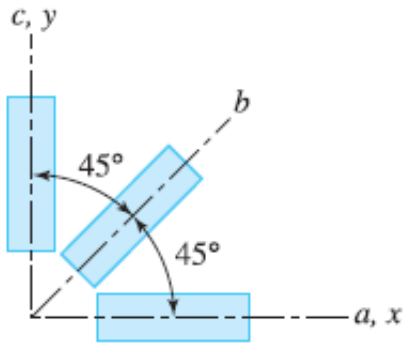
$$\frac{\gamma_\theta}{2} = -\frac{\varepsilon_x - \varepsilon_y}{2}\sin 2\theta + \frac{\gamma_{xy}}{2}\cos 2\theta$$

$$= \frac{520 + (250)}{2}\sin(2\times 26.57^\circ) + \frac{400}{2}\cos(2\times 26.57^\circ) = -447.77\mu$$

$$\Rightarrow \Delta\Phi = (\frac{\gamma_\theta}{2}) - (\frac{\gamma_{xy}}{2}) = -247.7\mu \Rightarrow \Delta\Phi = 247.7\mu(rad)\text{减少}$$

例 5

A 45° strain rosette mounted on the surface of an automobile frame gives the following readings: gage A, 600×10^{-6}; gage B, 300×10^{-6}; and gage C, -100×10^{-6}. Determine the principal strains and the maximum shear strains.



【清大動機】

【解析】

$\varepsilon_x = 600\mu = \varepsilon_a$ ， $\varepsilon_y = -100\mu = \varepsilon_c$

$$\varepsilon_\theta = \varepsilon_b = \frac{\varepsilon_x + \varepsilon_y}{2} + \frac{\varepsilon_x - \varepsilon_y}{2}\cos 2\theta + \frac{\gamma_{xy}}{2}\sin 2\theta = 300\mu$$

$$\Rightarrow \frac{600-100}{2} + \frac{600-(-100)}{2}\cos(2\times 45^\circ) + \frac{\gamma_{xy}}{2}\sin(2\times 45^\circ) = 300\mu \Rightarrow \gamma_{xy} = 100\mu$$

主應變 $\varepsilon_{1,2} = \frac{\varepsilon_x + \varepsilon_y}{2} \pm \sqrt{(\frac{\varepsilon_x - \varepsilon_y}{2})^2 + (\frac{\gamma_{xy}}{2})^2} \Rightarrow \varepsilon_1 = 603\mu, \varepsilon_2 = -103\mu$

最大剪應變 $\gamma_{\max} = \varepsilon_1 - \varepsilon_2 = 706\mu$

例 6

構件內某一點為平面應變狀態，已知$\varepsilon_x = 360\mu$，$\varepsilon_y = 750\mu$，$\varepsilon_2 = 197\mu$，試求(a) γ_{xy}，(b) ε_1，(c)同平面最大剪應變γ_{max}，(d)絕對最大剪應變$(\gamma_{max})_{abs}$，(e)最小正交應變之方向θ_{p2}。

【解析】

(b) $\varepsilon_1 = (\varepsilon_x + \varepsilon_y) - \varepsilon_2 = (360\mu + 750\mu) - 197\mu = 913\mu$

(c) $\gamma_{\max} = \varepsilon_1 - \varepsilon_2 = 913\mu - 197\mu = 716\mu$

(d) $\varepsilon_1 > 0$ ， $\varepsilon_2 > 0$ ， $(\gamma_{\max})_{abs} = \varepsilon_1 - 0 = 913\mu$

(a) $\dfrac{\gamma_{\max}}{2} = \sqrt{\left(\dfrac{\varepsilon_x - \varepsilon_y}{2}\right)^2 + \left(\dfrac{\gamma_{xy}}{2}\right)^2}$ ， $\dfrac{716\mu}{2} = \sqrt{\left(\dfrac{360\mu - 750\mu}{2}\right)^2 + \left(\dfrac{\gamma_{xy}}{2}\right)^2}$

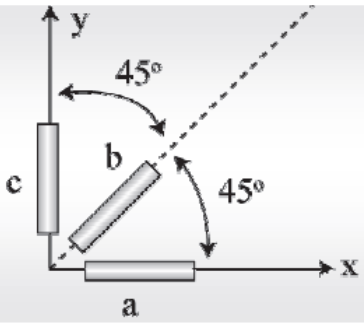
$\gamma_{xy} = \pm 600\ \mu$

(e) $\tan 2\theta_p = \dfrac{\gamma_{xy}}{\varepsilon_x - \varepsilon_y} = \dfrac{\pm 600\mu}{360\mu - 750\mu} = \dfrac{\pm 600\mu}{-390\mu}$

$2\theta_{p1} = \pm 123^\circ$ ， $\theta_{p1} = \pm 61.5^\circ$

例 7

實驗時在鋼材表面某點貼如圖所示 a,b,c 三組應變計。分別測得各應變值為 $\varepsilon_a = 0.0008$，$\varepsilon_b = 0.00096$，$\varepsilon_c = 0.0006$。求該點之主應變與最大剪應變。(96 年土木高考)



【解析】

解 如圖所示：$\varepsilon_x = \varepsilon_a = 0.0008 \quad \varepsilon_y = \varepsilon_c = 0.0006$

$$\varepsilon_b = \frac{\varepsilon_x + \varepsilon_y}{2} + \frac{\varepsilon_x - \varepsilon_y}{2}\cos(2\times 45°) + \frac{\gamma_{xy}}{2}\sin(2\times 45°) \Rightarrow \gamma_{xy} = 2\varepsilon_b - \varepsilon_x - \varepsilon_y$$

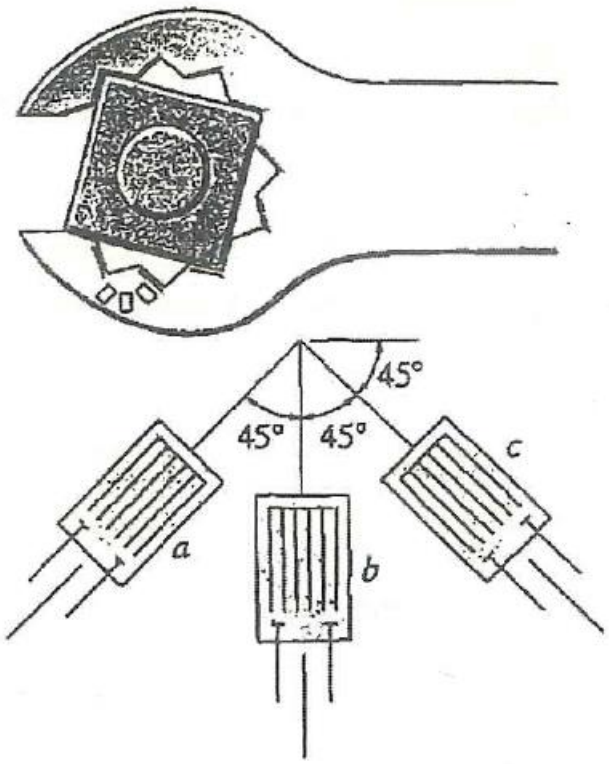
將 $\varepsilon_b = 0.00096$，$\varepsilon_x = \varepsilon_a = 0.0008$，$\varepsilon_y = \varepsilon_b = 0.0006$ 代入可得 $\gamma_{xy} = 5.2\times 10^{-4}$

$$\varepsilon_{1,2} = \frac{\varepsilon_x + \varepsilon_y}{2} \pm \sqrt{(\frac{\varepsilon_x - \varepsilon_y}{2})^2 + (\frac{\gamma_{xy}}{2})^2} \Rightarrow \varepsilon_1 = 9.786\times 10^{-4}\ ,\ \varepsilon_2 = 4.214\times 10^{-4}$$

$$\frac{\gamma_{max}}{2} = 2.786\times 10^{-4} \Rightarrow \gamma_{max} = 5.572\times 10^{-4}$$

例 8

(20%) The 45° strain rosette is mounted near the tooth of the wrench. The following readings are obtained for each gage: $\varepsilon_a = 800\mu$, $\varepsilon_b = 520\mu$, and $\varepsilon_c = -450\mu$. Determine the in-plane principal strains.



【99 中興機械】

【解析】

$$\varepsilon_y = 520\mu = \varepsilon_b$$

$$\varepsilon_\theta = \varepsilon_c = \frac{\varepsilon_x + \varepsilon_y}{2} + \frac{\varepsilon_x - \varepsilon_y}{2}\cos 2\theta + \frac{\gamma_{xy}}{2}\sin 2\theta = -450\mu$$

$$\Rightarrow \frac{\varepsilon_x + 520}{2} + \frac{\varepsilon_x - (520)}{2}\cos(2\times 45^\circ) + \frac{\gamma_{xy}}{2}\sin(2\times 45^\circ) = -450\mu \Rightarrow \varepsilon_x + \gamma_{xy} = -1420\mu$$

$$\varepsilon_\theta = \varepsilon_a = \frac{\varepsilon_x + \varepsilon_y}{2} + \frac{\varepsilon_x - \varepsilon_y}{2}\cos 2\theta + \frac{\gamma_{xy}}{2}\sin 2\theta = 800\mu$$

$$\Rightarrow \frac{\varepsilon_x + 520}{2} + \frac{\varepsilon_x - (520)}{2}\cos(-2\times 45^\circ) + \frac{\gamma_{xy}}{2}\sin(-2\times 45^\circ) = 800\mu \Rightarrow \varepsilon_x - \gamma_{xy} = 1080\mu$$

$$\gamma_{xy} = -1250\,\mu$$

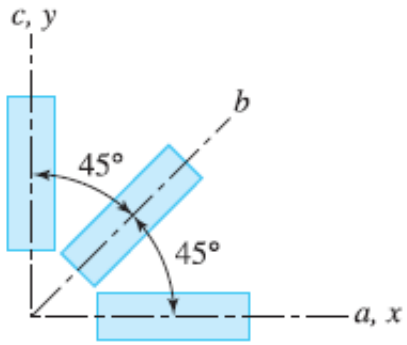
$$\varepsilon_x = -170\,\mu$$

主應變 $\varepsilon_{1,2}=\frac{\varepsilon_x+\varepsilon_y}{2}\pm\sqrt{(\frac{\varepsilon_x-\varepsilon_y}{2})^2+(\frac{\gamma_{xy}}{2})^2}\Rightarrow\varepsilon_1=888.9\mu,\varepsilon_2=-538.9\mu$

最大剪應變 $\gamma_{\max}=\varepsilon_1-\varepsilon_2=1427.8\mu$

例 9

The strains determined by use of the rosette shown are $\varepsilon_a = 500\times10^{-6}$, $\varepsilon_b = -120\times10^{-6}$, $\varepsilon_c = 100\times10^{-6}$. Determine the shearing strain γ_{xy} , the in-plane principal strains , the in-plane maximum shear strain.



【成大土木】

【解析】

$\varepsilon_x = 500\mu = \varepsilon_a$, $\varepsilon_y = 100\mu = \varepsilon_c$

$$\varepsilon_\theta = \varepsilon_b = \frac{\varepsilon_x+\varepsilon_y}{2} + \frac{\varepsilon_x-\varepsilon_y}{2}\cos 2\theta + \frac{\gamma_{xy}}{2}\sin 2\theta = -120\mu$$

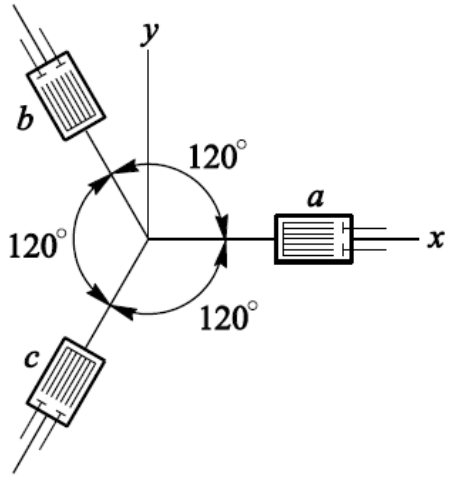
$$\Rightarrow \frac{500+100}{2} + \frac{500-100}{2}\cos(2\times45^\circ) + \frac{\gamma_{xy}}{2}\sin(2\times45^\circ) = -120\mu \Rightarrow \gamma_{xy} = -840\mu$$

主應變 $\varepsilon_{1,2} = \frac{\varepsilon_x+\varepsilon_y}{2} \pm \sqrt{(\frac{\varepsilon_x-\varepsilon_y}{2})^2 + (\frac{\gamma_{xy}}{2})^2} \Rightarrow \varepsilon_1 = 765.2\mu, \varepsilon_2 = -165.2\mu$

最大剪應變 $\gamma_{\max} = \varepsilon_1 - \varepsilon_2 = 930.4\mu$

例 10

構件內某一點利用應變計所受之平面應變如下：$\varepsilon_a = -555\mu$，$\varepsilon_b = 925\mu$，$\varepsilon_c = 740\mu$。試求(a)主應變，(b)同平面最大剪應變。



【解析】

$\varepsilon_x = -550\mu = \varepsilon_a$

$$\varepsilon_{120^\circ} = \varepsilon_b = \frac{-550+\varepsilon_y}{2} + \frac{-550-\varepsilon_y}{2}\cos 2\times 120^\circ + \frac{\gamma_{xy}}{2}\sin 2\times 120^\circ = -925\mu$$

$$\varepsilon_{-120^\circ} = \varepsilon_c = \frac{-550+\varepsilon_y}{2} + \frac{-550-\varepsilon_y}{2}\cos 2\times -120^\circ + \frac{\gamma_{xy}}{2}\sin 2\times -120^\circ = 740\mu$$

上兩式解聯立的得

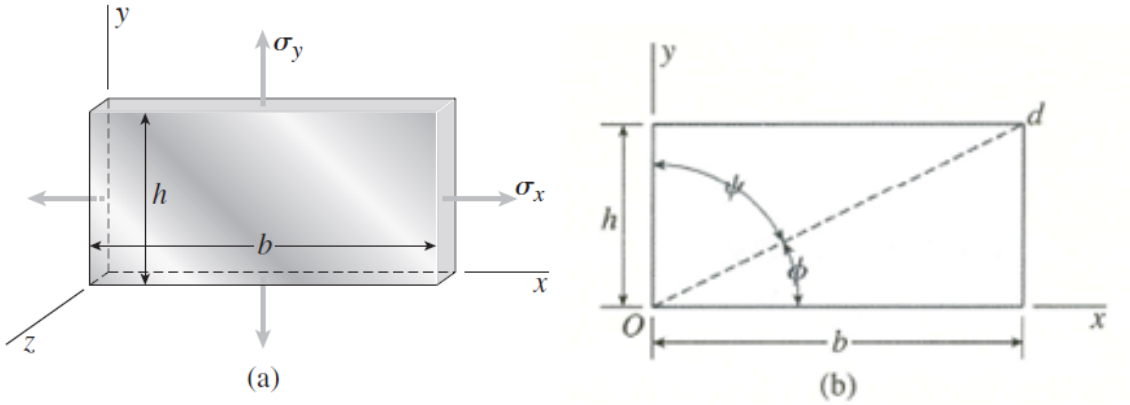
$\varepsilon_y = 1295\mu$且$\gamma_{xy} = -210\mu$

主應變 $\varepsilon_{1,2} = \frac{\varepsilon_x+\varepsilon_y}{2} \pm \sqrt{(\frac{\varepsilon_x-\varepsilon_y}{2})^2 + (\frac{\gamma_{xy}}{2})^2} \Rightarrow \varepsilon_1 = 1301\mu, \varepsilon_2 = -561\mu$

最大剪應變 $\gamma_{\max} = \varepsilon_1 - \varepsilon_2 = 1862\mu$

例 11

A thin rectangular plate in biaxial stress is subjected to stresses x and y, as shown in part (a) of the figure on the next page. The width and height of the plate are $b = 200\ \text{mm}$ and $h = 100\ \text{mm}$, respectively. Measurements show that the normal strains in the x and y directions are $\varepsilon_x = 195 \times 10^{-6}$ and $\varepsilon_y = -125 \times 10^{-6}$, respectively. With reference to part (b) of the figure, which shows a two-dimensional view of the plate, determine the following quantities: (a) the increase Δd in the length of diagonal Od; (b) the change $\Delta\phi$ in the angle Between diagonal Od and the x axis; and (c) the change $\Delta\psi$ in the angle between diagonal Od and the y axis.



【清大動機類題】

【中文翻譯】

雙軸向應力薄短形板受應力 σ_x 及 σ_y，如圖 (a) 所示，寬 $b = 200\ \text{mm}$ 及高度 $h = 100\ \text{mm}$。以量度得知，x 及 y 方向之正向應變分別為 $\varepsilon_x = 195 \times 10^{-6}$ 及 $\varepsilon_y = -125 \times 10^{-6}$。參照圖 (b)，係板的二維視圖，求如下的量：(a) 對角線 Od 的增長 Δd；(b) x 軸與對角線 Od 之間的角度之變化 $\Delta\phi$；(c) y 軸與對角線 Od 之間的角度之變化 $\Delta\psi$。

【解析】

(a)

$\mathbf{b = 200\ mm,\ h = 100\ mm,\ \varepsilon_x = 195 \times 10^{-6}}$

$\mathbf{\varepsilon_y = -125 \times 10^{-6},\ \gamma_{xy} = 0}$

$$\phi = \arctan\frac{h}{b} = 26.57°$$

$$L_d = \sqrt{b^2 + h^2} = \mathbf{223.61\ mm}$$

$$\varepsilon_\theta = \varepsilon_b = \frac{\varepsilon_x + \varepsilon_y}{2} + \frac{\varepsilon_x - \varepsilon_y}{2}\cos 2\theta + \frac{\gamma_{xy}}{2}\sin 2\theta = \Delta d$$

$$\Rightarrow \frac{200 + (-195)}{2} + \frac{200 - (-195)}{2}\cos(2 \times 26.57^\circ) + \frac{\gamma_{xy}}{2}\sin(2 \times 26.57^\circ) = \Delta d \Rightarrow L_d = 0.02929mm$$

(b)

$$\frac{\gamma_\theta}{2} = -\frac{\varepsilon_x - \varepsilon_y}{2}\sin 2\theta + \frac{\gamma_{xy}}{2}\cos 2\theta$$

$$\frac{200 - (-195)}{2}\sin(2 \times 26.57^\circ) + \frac{\gamma_{xy}}{2}\cos(2 \times 26.57^\circ) = -128\mu \Rightarrow \Delta\Phi = 128\mu(rad)\text{減少}$$

$$\Rightarrow \Delta\Phi = (\frac{\gamma_\theta}{2}) - (\frac{\gamma_{xy}}{2}) = -128\mu \Rightarrow \Delta\Phi = 128\mu(rad)\text{減少}$$

(c)

$\Delta\psi = 128\mu(rad)$增加